RINGS AND MODULES

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Contents

1 RINGS 1
- 1.1 Properties and Maps . . . 1
  - 1.1.1 Polynomials . . . 3
- 1.2 Ideals . . . 4
- 1.3 Other Rings . . . 6
- 1.4 Quotient Rings and Isomorphism Theorems . . . 7
- 1.5 Prime and Maximal Ideals . . . 9
  - 1.5.1 Jacobson Radical and Nilradical . . . 11
- 1.6 Product of Rings . . . 12
  - 1.6.1 Idempotents . . . 13
- 1.7 Generalizing Properties of Integers . . . 14
- 1.8 Localization . . . 19
  - 1.8.1 Local Ring . . . 22
- 1.9 Polynomials and Gauss Lemma . . . 22

Index 25

Chapter 1

RINGS

January 19th.

Of course, we begin with the definition of a ring.

Definition 1.1. A *ring* is a triple $(R, +, \cdot)$ where R is a set, and $+$ and $\cdot$ are binary operations on R such that the following axioms are satisfied:

- $(R, +)$ is an abelian group. The identity element of this group is denoted by 0_R, and the (additive) inverse of an element $a \in R$ is denoted by $-a$.
- The property of *associativity* of $\cdot$ holds; i.e., for all $a, b, c \in R$, we have $(a \cdot b) \cdot c = a \cdot (b \cdot c)$.
- The property of *distributivity* of $\cdot$ over $+$ holds; i.e., for all $a, b, c \in R$, we have

$$a \cdot (b + c) = a \cdot b + a \cdot c, \tag{1.1}$$
$$(a + b) \cdot c = a \cdot c + b \cdot c. \tag{1.2}$$

Rings may be written simply as R instead of the triple. The ring R is termed a *ring with unity* if there exists an element $1_R \in R$ such that for all $a \in R$, we have $1_R \cdot a = a \cdot 1_R = a$. Some examples of rings with unity include $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$, $M_n(\mathbb{R})$ with the usual addition and multiplication. A ring R is said to be a *commutative ring* if for all $a, b \in R$, we have $a \cdot b = b \cdot a$. Examples of commutative rings include $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$, but $M_n(\mathbb{R})$ is not commutative for $n \geq 2$. Lastly, a commutative ring R with unity is termed a *field* if every non-zero element of R has a multiplicative inverse; i.e., for every $a \in R \setminus \{0_R\}$, there exists an element $b \in R$ such that $a \cdot b = b \cdot a = 1_R$. Examples of fields include $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$, but $\mathbb{Z}$ is not a field.

Example of rings without unity include $2\mathbb{Z}$ with the usual addition and multiplication, and the set of all continuous functions from $\mathbb{R}$ to $\mathbb{R}$ that vanish at 0, with the usual addition and multiplication of functions. Another class of rings we previously studied was $\mathbb{Z}/n\mathbb{Z}$ for $n \geq 2$, with the usual addition and multiplication modulo n. This ring has unity, but is a field if and only if n is prime.

Definition 1.2. Let R be a ring with unity. An element $a \in R$ is called a *unit* if there exists an element $b \in R$ such that $a \cdot b = b \cdot a = 1_R$.

For example, in the ring $\mathbb{Z}/n\mathbb{Z}$, an element $\bar{a}$ is a unit if and only if $\gcd(a, n) = 1$. The set of all units in a ring R with unity is denoted by $R^\times$. It can be easily verified that $(R^\times, \cdot)$ is an abelian group.

1.1 Properties and Maps

Some basic properties may be inferred.

Proposition 1.3. *Let R be a ring with unity. Then,*

- *1_R is the unique multiplicative identity in R.*
- *$1_R \cdot 0_R = 0_R$. In general, $a \cdot 0_R = 0_R$ for all $a \in R$.*
- *$-1_R \cdot a = -a$ for all $a \in R$.*

Proof.
- This is left as an exercise to the reader.
- $1_R \cdot 0_R = 1_R$ is trivial since 1_R is the multiplicative identity. For the general case, let $a \in R$. Then,

$$a \cdot 0_R = a \cdot (0_R + 0_R) = a \cdot 0_R + a \cdot 0_R \implies a \cdot 0_R = 0_R \tag{1.3}$$

by the addition of $-(a \cdot 0_R)$ on both sides.

- Let $a \in R$. Then,

$$(-1_R \cdot a) + a = (-1_R + 1_R) \cdot a = 0_R \implies -1_R \cdot a = -a. \tag{1.4}$$

■

The subscript R in 0_R and 1_R may be dropped when the context is clear. We move on to some special maps.

Definition 1.4. A *ring homomorphism* is a map $\varphi : (R, +, \cdot) \to (S, \oplus, \odot)$ between two rings such that for all $a, b \in R$, we have

$$\varphi(a + b) = \varphi(a) \oplus \varphi(b), \quad \varphi(a \cdot b) = \varphi(a) \odot \varphi(b). \tag{1.5}$$

Most of the time, we shall drop $\oplus$ and $\odot$ when the context is clear. Some examples of ring homomorphisms include the map $\varphi : \mathbb{Z} \to \mathbb{Z}/n\mathbb{Z}$ defined by $\varphi(a) = \bar{a}$ for all $a \in \mathbb{Z}$, and the inclusion map from $\mathbb{Z}$ to $\mathbb{Q}$. Non-examples include $n \mapsto -n$ from $\mathbb{Z}$ to $\mathbb{Z}$, and the determinant map from $M_n(\mathbb{R})$ to $\mathbb{R}$.

Let $(\mathbb{Z} \times \mathbb{Z}, +, \cdot)$ be the ring where addition and multiplication are defined component-wise. Then the map $Z \to \mathbb{Z} \times \mathbb{Z}$ defined by $a \mapsto (a, 0)$ is a ring homomorphism since it preserves both addition and multiplication. However, the unity of $\mathbb{Z}$ is mapped to $(1, 0)$, which is not the unity of $\mathbb{Z} \times \mathbb{Z}$. Thus, ring homomorphisms need not map unity to unity.

Definition 1.5. Let R be a ring with $S \subseteq R$ a subset. Then, S is called a *subring* of R if $(S, +, \cdot)$ is itself a ring with the operations inherited from R.

Again, even if R has unity, a subring S need not have the same unity as R or even a unity at all.

January 23rd.

Definition 1.6. A ring homomorphism $\varphi : R \to S$ is termed a *ring monomorphism* if it is injective, a *ring epimorphism* if it is surjective, and a *ring isomorphism* if it is bijective. If there exists a ring isomorphism from R to S, then R and S are said to be *isomorphic*, denoted by $R \cong S$.

Note that if $\varphi : R \to S$ is bijective, then its inverse $\varphi^{-1} : S \to R$ is a ring homomorphism. We look at some examples of rings and mappings.

Example 1.7. Let X be any set and let $R := \{f : X \to \mathbb{R}\}$ be the set of all functions from X to $\mathbb{R}$. Then, $(R, +, \cdot)$ is a ring where addition and multiplication are defined pointwise; i.e., for all $f, g \in R$ and $x \in X$, $(f + g)(x) := f(x) + g(x)$ and $(f \cdot g)(x) := f(x) \cdot g(x)$. The additive identity is the zero function $0 : X \to \mathbb{R}$ defined by $0(x) = 0$ for all $x \in X$, and the multiplicative identity is the constant function $1 : X \to \mathbb{R}$ defined by $1(x) = 1$ for all $x \in X$. It is easy to verify that all ring axioms are

satisfied. Moreover, this ring is commutative and has unity. Note that $\mathbb{R}$ can be replaced by any ring S to form the ring of functions from X to S. In such a case, R is a (commutative) ring with unity if and only if S is a (commutative) ring with unity.
In the special case that $X = \{1, 2, \ldots, n\}$ for some $n \in \mathbb{N}$, the ring R is isomorphic to the ring $(\mathbb{R}^n, +, \cdot)$ where addition and multiplication are defined component-wise. The isomorphism $\varphi : R \to \mathbb{R}^n$ is given by $\varphi(f) = (f(1), f(2), \ldots, f(n))$ for all $f \in R$.

Example 1.8. Continuing from the previous example, let $X = [a, b]$. Note that the R in this case is the set of all functions from the interval $[a, b]$ to $\mathbb{R}$, which is not a very manageable set. Thus, we may consider the subset $C([a, b], \mathbb{R}) \subseteq R$ consisting of all continuous functions from $[a, b]$ to $\mathbb{R}$. It is easy to verify that $C([a, b], \mathbb{R})$ is a subring of R. Similarly, one defined $C^n([a, b], \mathbb{R})$ to be the set of all n-times continuously differentiable functions from $[a, b]$ to $\mathbb{R}$, and $C^\infty([a, b], \mathbb{R})$ to be the set of all infinitely differentiable functions from $[a, b]$ to $\mathbb{R}$. Both of these are subrings of R as well.

Example 1.9. The set $\mathbb{Z}[i] := \{a + bi : a, b \in \mathbb{Z}\}$ is a subring of the field $\mathbb{C}$. It is easy to verify that $\mathbb{Z}[i]$ is a ring with unity, but it is not a field since, for example, the element $1 + i$ does not have a multiplicative inverse in $\mathbb{Z}[i]$. Note that there is a natural bijection $\varphi : \mathbb{Z}[i] \to \mathbb{Z}^2$ defined by $\varphi(a + bi) = (a, b)$ for all $a + bi \in \mathbb{Z}[i]$, where $\mathbb{Z}^2$ has component-wise addition and multiplication. However, this map is not a ring isomorphism since it does not preserve multiplication; for example, $\varphi(i \cdot i) = \varphi(-1) = (-1, 0)$, but $\varphi(i) \cdot \varphi(i) = (0, 1) \cdot (0, 1) = (0, 1)$.

1.1.1 Polynomials

Let R be a ring. The polynomial ring in the variable x with coefficients from R is defined as follows:

Definition 1.10. The *polynomial ring* $R[x]$ is defined as

$$R[x] := \{f : \mathbb{N}_0 \to R \mid f(n) = 0 \text{ for all but finitely many } n \in \mathbb{N}_0\}. \tag{1.6}$$

The elements of $R[x]$ are called *polynomials* in the variable x with coefficients from R. For $f, g \in R[x]$ and $n \in \mathbb{N}_0$, addition is defined as

$$(f + g)(n) := f(n) + g(n) \quad \text{for all } n \in \mathbb{N}_0, \tag{1.7}$$

and multiplication is defined as

$$(f \cdot g)(n) := \sum_{k=0}^{n} f(k) \cdot g(n - k) \quad \text{for all } n \in \mathbb{N}_0. \tag{1.8}$$

Alternatively, a polynomial $f \in R[x]$ may be expressed in the form

$$f(x) = a_0 + a_1 x + a_2 x^2 + \cdots + a_n x^n, \tag{1.9}$$

where $a_i = f(i)$ for all $0 \leq i \leq n$ and $f(k) = 0$ for all $k > n$. For $0 \neq f \in R[x]$ as above with $a_n \neq 0_R$, the integer n is called the *degree* of f, denoted by $\deg(f)$. The degree of the zero polynomial is usually left undefined, or changed upon convention. Also note that $f \cdot g \in R[x]$ since $f \cdot g(k) = 0_R$ for all $k > \deg(f) + \deg(g)$.

Proposition 1.11. *For a ring R, the polynomial ring $R[x]$ is, indeed, a ring with unity under the operations defined above. If R is commutative, then so is $R[x]$. The map $\iota : R \to R[x]$ defined by $\iota(a) = f_a$ where $f_a(0) = a$ and $f_a(n) = 0_R$ for all $n \geq 1$ is a ring monomorphism.*

Proof. That $(R[x], +)$ forms an abelian group is clear. The associativity of multiplication is verified as

follows: let $f, g, h \in R[x]$ and $n \in \mathbb{N}_0$. Then,

$$\begin{aligned}((f \cdot g) \cdot h)(n) &= \sum_{k=0}^{n} (f \cdot g)(k) \cdot h(n-k) = \sum_{k=0}^{n} \left(\sum_{j=0}^{k} f(j) \cdot g(k-j) \right) \cdot h(n-k) \\ &= \sum_{j=0}^{n} f(j) \cdot \left(\sum_{k=j}^{n} g(k-j) \cdot h(n-k) \right) = \sum_{j=0}^{n} f(j) \cdot (g \cdot h)(n-j) = (f \cdot (g \cdot h))(n). \quad (1.10)\end{aligned}$$

The distributive properties follow similarly. The unity in $R[x]$ is the polynomial $1_{R[x]}$ defined by $1_{R[x]}(0) = 1_R$ and $1_{R[x]}(n) = 0_R$ for all $n \geq 1$. Finally, it is easy to verify that ι is a ring homomorphism, and it is injective since $\iota(a) = \iota(b)$ implies that $a = b$. ■

With $R[x]$ established as a ring, we may consider a higher level of abstraction, by considering polynomials over this polynomial ring itself; that is, $(R[x])[y]$. Elements of this ring look like

$$f(x, y) = a_{00} + a_{10}x + a_{01}y + a_{20}x^2 + a_{11}xy + a_{02}y^2 + \cdots + a_{mn}x^m y^n, \quad (1.11)$$

where $a_{ij} \in R$ for all $i, j \geq 0$ and $a_{ij} = 0_R$ for all but finitely many pairs (i, j). We have already shown that $R[x]$ is a ring, so it follows that $(R[x])[y]$ is also a ring. This ring is usually denoted by $R[x, y]$. For $f \in R[x, y]$ as above with $a_{mn} \neq 0_R$, the degree of f is defined as $\deg(f) = m + n$. Similarly, one may define $R[x_1, x_2, \ldots, x_n]$ for any $n \in \mathbb{N}$. For a countable number of indeterminates, one may define $R[x_1, x_2, x_3, \ldots]$ as the union $\bigcup_{n=1}^{\infty} R[x_1, x_2, \ldots, x_n]$.

Example 1.12. Let $e \in \mathbb{R}$ be the Euler's number (or any transcendental number). Then $\mathbb{Z}[e] \subseteq \mathbb{C}$ is the smallest subring of $\mathbb{C}$ containing both $\mathbb{Z}$ and e. Here, $\mathbb{Z}[e]$ consists of all polynomials in e with integer coefficients; i.e., all elements of the form $a_0 + a_1 e + a_2 e^2 + \cdots + a_n e^n$ where $n \geq 0$ and $a_i \in \mathbb{Z}$. Since e is transcendental, there are no non-trivial polynomial relations among the powers of e with integer coefficients. Thus, the map $\varphi : \mathbb{Z}[x] \to \mathbb{Z}[e]$ defined by $\varphi(f) = f(e)$ for all $f \in \mathbb{Z}[x]$ is a ring isomorphism.

1.2 Ideals

Definition 1.13. Let R be a commutative ring with unity. A subset $I \subseteq R$ is called an *ideal* of R if the following conditions hold:

- for all $a, b \in I$, we have $a + b \in I$,
- for all $a \in I$ and $r \in R$, we have $r \cdot a \in I$.

Note that the first condition implies that $(I, +)$ is a subgroup of $(R, +)$. Some examples of ideals include the set $\{0_R\}$, the ring R itself, and the set $n\mathbb{Z} = \{nk : k \in \mathbb{Z}\}$ for any $n \in \mathbb{Z}_{\geq 0}$ as an ideal of the ring $\mathbb{Z}$. A non-example is $\mathbb{Z}$ in $\mathbb{R}$; it is a subring, but not an ideal since, for example, $1 \in \mathbb{Z}$ but $\pi \cdot 1 = \pi \notin \mathbb{Z}$. Note that if $1_R \in I$, then $I = R$.

Example 1.14. Let us look at ideals of $\mathbb{R}$. Trivially, $\{0\}$ and $\mathbb{R}$ are ideals of $\mathbb{R}$. We claim that these are the only ideals of $\mathbb{R}$. To see this, let I be any ideal of $\mathbb{R}$ such that $I \neq \{0\}$. Then, there exists some $a \in I$ such that $a \neq 0$. Since $\mathbb{R}$ is a field, a has a multiplicative inverse $a^{-1} \in \mathbb{R}$. Thus, $1 = a^{-1} \cdot a \in I$, which implies that $I = \mathbb{R}$. In fact, this argument shows that in any field, the only ideals are the zero ideal and the field itself.

Example 1.15. We examine ideals of $\mathbb{Z}$. From group theory, we know that every subgroup of $(\mathbb{Z}, +)$ is of the form $n\mathbb{Z}$ for some $n \in \mathbb{Z}_{\geq 0}$, so $n\mathbb{Z}$ are the only candidates for ideals of $\mathbb{Z}$. In fact, each $n\mathbb{Z}$ is an ideal of $\mathbb{Z}$ since for all $a, b \in n\mathbb{Z}$, we have $a + b \in n\mathbb{Z}$, and for all $a \in n\mathbb{Z}$ and $r \in \mathbb{Z}$, we have $r \cdot a \in n\mathbb{Z}$. Thus, the ideals of $\mathbb{Z}$ are precisely the sets $n\mathbb{Z}$ for $n \in \mathbb{Z}_{\geq 0}$, and $\mathbb{Z}$.

Proposition 1.16. *Let $f : R \to S$ be a ring homomorphism between two commutative rings with unity. Then, the* kernel *of f, defined as*

$$\ker f := \{a \in R \mid f(a) = 0_S\}, \tag{1.12}$$

is an ideal of R. Moreover, f is a ring monomorphism if and only if $\ker f = \{0_R\}$.

Proof. Let $a, b \in \ker f$ and $r \in R$. Then,

$$f(a+b) = f(a) + f(b) = 0_S + 0_S = 0_S, \tag{1.13}$$

so $a + b \in \ker f$. Also,

$$f(r \cdot a) = f(r) \cdot f(a) = f(r) \cdot 0_S = 0_S, \tag{1.14}$$

so $r \cdot a \in \ker f$. Thus, $\ker f$ is an ideal of R.

Now, suppose that f is a ring monomorphism. Let $a \in \ker f$. Then, $f(a) = 0_S = f(0_R)$. Since f is injective, we have $a = 0_R$, so $\ker f = \{0_R\}$. Conversely, suppose that $\ker f = \{0_R\}$. Let $a, b \in R$ such that $f(a) = f(b)$. Then,

$$f(a-b) = f(a) - f(b) = 0_S, \tag{1.15}$$

so $a - b \in \ker f$. Thus, $a - b = 0_R$, which implies that $a = b$. Therefore, f is injective. ■

January 24th.

Let R be a ring with unity and R_i be a colelction of subrings of R containing the nuity. Then $\bigcap_i R_i$ is also a subring of R containing the unity. If I_j is a collection of ideals of R, then $\bigcap_j I_j$ is also an ideal of R. Thus, given any subset $S \subseteq R$, we may define the ideal generated.

Definition 1.17. Let R be a commutative ring with unity and $I \subseteq R$ be an ideal. Let $S \subseteq I$ be a set. We say S is a *generating set* of I if I is the smallest ideal containing S.

Proposition 1.18. *Let R be a commutative ring with unity and $S \subseteq R$ be any subset. Then, the ideal generated by S, denoted by (S), is given by*

$$(S) = \left\{ \sum_{i=1}^{n} r_i s_i : n \geq 0, r_i \in R, s_i \in S \text{ for all } 1 \leq i \leq n \right\}. \tag{1.16}$$

Proof. Let $S \subseteq I$, a subset of an ideal. We claim that $(S) \subseteq I$. Let $\alpha \in I$. THen, $\alpha = r_1 x_1 + \cdots + r_n x_n$ for some $n \geq 0$, $r_i \in R$ and $x_i \in S$ for all $1 \leq i \leq n$. Since I is an ideal, we have $r_i x_i \in I$ for all $1 \leq i \leq n$, and thus $\alpha \in I$. Therefore, $(S) \subseteq I$. ■

With this, we introduce the notation that if $\{x_1, \ldots, x_n\} \subseteq R$, then $I = (x_1, \ldots, x_n) = Rx_1 + \cdots + Rx_n$ is the ideal generated by $x_1, \ldots, x_n$. Let us look at some exmaples.

Example 1.19. In the ring $\mathbb{Z}$, $(2, 3) = \mathbb{Z}$ since $1 = 3 - 1 \cdot 2 \in (2, 3)$. More generally, for any $a, b \in \mathbb{Z}$, we have $(a, b) = \mathbb{Z}$ if and only if $\gcd(a, b) = 1$. Moreover, in $\mathbb{Z}$, every ideal can be generated by a single element; i.e., every ideal is of the form (n) for some $n \in \mathbb{Z}_{\geq 0}$.

Example 1.20. In $\mathbb{Z}[x]$, the ideal $(2, x)$ consists of all polynomials with integer coefficients where the constant term is even. That is, $(2, x) = 2\mathbb{Z}[x] + x\mathbb{Z}[x]$.

Also note that a union of ideals need not be an ideal. For example, in $\mathbb{Z}$, the sets $2\mathbb{Z}$ and $3\mathbb{Z}$ are ideals, but their union $2\mathbb{Z} \cup 3\mathbb{Z}$ is not an ideal. This, however, calls for a more general construction.

Definition 1.21. Let R be a commutative ring with unity. If I_1, I_2 are two ideals, we then define their sum as $I_1 + I_2 = (I_1 \cup I_2)$.

It is easy to verify that $I_1 + I_2 = \{a + b \mid a \in I_1, b \in I_2\}$. This definition may be extended to a finite number of ideals in the obvious way.

1.3 Other Rings

Definition 1.22. Let G be a group and k be a field. We define $R[G]$ to be the set of all functions $f : G \to k$ such that $f(x) = 0$ for all but finitely many $x \in G$. Addition is defined pointwise as

$$(f + g)(x) = f(x) + g(x) \quad \text{for all } x \in G, \tag{1.17}$$

and multiplication is defined as

$$(f \cdot g)(x) = \sum_{yz=x} f(y)g(z) = \sum_{y \in G} f(xy^{-1})g(y) \quad \text{for all } x \in G. \tag{1.18}$$

The ring $R[G]$ is called the *group ring* of G over k.

If G is a finite group with $G = \{e, x_2, \ldots, x_n\}$ then $R[G] = \{a_1 e + a_2 x_2 + \cdots + a_n x_n \mid a_i \in \mathbb{C}\}$. Verify that $R[G]$ is a ring with unity under the operations defined above. This ring, however, may not be commutative.

Definition 1.23. Let R be a ring and $x \in R$. x is termed *nilpotent* if there exists some $n \in \mathbb{N}$ such that $x^n = 0$. If R is commutative, $x \in R$ is called a *zero divisor* if there exists some $y \in R \setminus \{0\}$ such that $x \cdot y = 0$.

Note that nilpotents are zero divisors in a commutative ring, but the converse need not be true. For example, in the ring $\mathbb{Z}/6\mathbb{Z}$, the element $\bar{2}$ is a zero divisor since $\bar{2} \cdot \bar{3} = \bar{0}$, but it is not nilpotent since $\bar{2}^n \neq \bar{0}$ for all $n \geq 1$.

Definition 1.24. A commutative ring with unity R is called a *reduced ring* if it has no non-zero nilpotent elements. It is called an *integral domain* if it has no non-zero zero divisors.

Proposition 1.25. *Let R be an integral domain. Then, if $x, y \in R$ are such that $x \cdot y = 0$, then either $x = 0$ or $y = 0$.*

Proof. If $x \neq 0$, then since R is an integral domain, x is not a zero divisor. Thus, y must be 0. Similarly, if $y \neq 0$, then x must be 0. ■

Proposition 1.26. *Every integral domain is a reduced ring.*

Proof. Let R be an integral domain and let $x \in R$ be nilpotent. Then, there exists some $n \in \mathbb{N}$ such that $x^n = 0$. If $x \neq 0$, then since R is an integral domain, x is not a zero divisor. However, this contradicts the fact that $x^n = 0$. Thus, we must have $x = 0$, so R has no non-zero nilpotent elements. ■

January 29th.

In an integral domain R, if $ab = ac$ for some $a, b, c \in R$, then either $a = 0$ or $b = c$. Let us look at some examples of integral domains.

Example 1.27. The ring $\mathbb{Z}$ is an integral domain since it has no non-zero zero divisors. Similarly, the rings $\mathbb{Q}$, $\mathbb{R}$, and $\mathbb{C}$ are integral domains as well. More generally, any field is an integral domain. Moreover, $\mathbb{Z}/n\mathbb{Z}$ is an integral domain if and only if n is prime.

Example 1.28. Note that R is an integral domain if and only if $R[x]$ is an integral domain.

Another small result is as follows: if R is an integral domain and R' is a subring of R containing the unity, then R' is also an integral domain. Some non-examples of integral domains include $\mathbb{Z}^2$, $C[0,1]$, $C^\infty[0,1]$, etc.

1.4 Quotient Rings and Isomorphism Theorems

From here on, we shall assume that all rings are commutative with unity unless otherwise stated.

Definition 1.29. Let R be a ring and I be an ideal of R. The *quotient ring* R/I is defined as the set of all cosets of I in R; i.e.,

$$R/I := \{a + I : a \in R\}. \tag{1.19}$$

Addition and multiplication in R/I are defined as follows: for all $a, b \in R$,

$$(a + I) + (b + I) := (a + b) + I, \tag{1.20}$$
$$(a + I) \cdot (b + I) := (a \cdot b) + I. \tag{1.21}$$

Of course, we must verify that these operations are well-defined. Note that I is a normal subgroup of $(R, +)$ since R is abelian under addition, so R/I is an abelian group under addition. We verify that multiplication is well-defined as follows: let $a, a', b, b' \in R$ such that $a + I = a' + I$ and $b + I = b' + I$. Then, there exist $i_1, i_2 \in I$ such that $a' = a + i_1$ and $b' = b + i_2$. Thus,

$$\begin{aligned}(a' \cdot b') + I &= ((a + i_1) \cdot (b + i_2)) + I = (a \cdot b + a \cdot i_2 + i_1 \cdot b + i_1 \cdot i_2) + I \\ &= (a \cdot b) + I,\end{aligned} \tag{1.22}$$

since $a \cdot i_2, i_1 \cdot b, i_1 \cdot i_2 \in I$. Therefore, multiplication is well-defined. Moreover, it is easy to verify that R/I is a ring with unity under these operations, where the additive identity is $0 + I$ and the multiplicative identity is $1 + I$. One also has the *quotient map* naturally defined as

$$q : R \to R/I, \quad q(a) = a + I \quad \text{for all } a \in R. \tag{1.23}$$

It is easy to verify that q is a ring epimorphism with kernel I. The most common example of a quotient ring is $\mathbb{Z}/n\mathbb{Z}$, which is isomorphic to the quotient ring $\mathbb{Z}/(n\mathbb{Z})$.

Example 1.30. In the ring $\mathbb{Q}[x]$, let $I = (x^2 - 2) = (x^2 - 2)\mathbb{Q}$. Then, the quotient $\mathbb{Q}[x]/(x^2 - 2)$ is indeed a quotient ring. It is also a field, and may be written as $\mathbb{Q}[\sqrt{2}]$. However, the quotient ring $\mathbb{R}[x]/(x^2 - 2)$ is not an integral domain since $(x - \sqrt{2} + I)(x + \sqrt{2} + I) = x^2 - 2 + I = I$.

We are now fit to show the isomorphism theorems for rings.

Theorem 1.31 (The *first isomorphism theorem* for rings)**.** *Let $\varphi : R \to S$ be a ring homomorphism. Let $I = \ker \varphi$. Then there exists a unique ring monomorphism $\overline{\varphi} : R/I \to S$ such that $\varphi = \overline{\varphi} \circ q$, where $q : R \to R/I$ is the quotient map. Moreover, if φ is surjective, then $\overline{\varphi}$ is a ring isomorphism.*

Proof. Define the map $\overline{\varphi} : R/I \to S$ as follows: for all $a + I \in R/I$, let

$$\overline{\varphi}(a + I) = \varphi(a). \tag{1.24}$$

We must verify that this map is well-defined. Let $a, b \in R$ such that $a + I = b + I$. Then, there exists some $i \in I$ such that $b = a + i$. Thus,

$$\varphi(b) = \varphi(a + i) = \varphi(a) + \varphi(i) = \varphi(a) + 0_S = \varphi(a), \tag{1.25}$$

so $\overline{\varphi}$ is well-defined. It is easy to verify that $\overline{\varphi}$ is a ring homomorphism. Also, for all $a \in R$,

$$(\overline{\varphi} \circ q)(a) = \overline{\varphi}(a + I) = \varphi(a), \tag{1.26}$$

so $\varphi = \overline{\varphi} \circ q$.

Now, suppose that $\overline{\varphi}(a + I) = 0_S$ for some $a + I \in R/I$. Then, $\varphi(a) = 0_S$, so $a \in I$. Thus, $a + I = I$, which is the additive identity in R/I. Therefore, $\overline{\varphi}$ is injective.

Finally, if φ is surjective, then for any $s \in S$, there exists some $a \in R$ such that $\varphi(a) = s$. Thus,

$$\overline{\varphi}(a + I) = \varphi(a) = s, \tag{1.27}$$

so $\overline{\varphi}$ is surjective as well. Therefore, $\overline{\varphi}$ is a ring isomorphism. ■

Proposition 1.32. *Let R be a ring and I be an ideal of R. Then there is a bijection between the set of all ideals of R containing I and the set of all ideals of the quotient ring R/I.*

Proof. We make use of the quotient map $q : R \to R/I$. Let J be an ideal of R such that $I \subseteq J$. The bijection is given by sending J to $J/I := q(J) = \{a + I : a \in J\}$, and sending K, an ideal of R/I, to $q^{-1}(K) = \{a \in R : q(a) \in K\}$. We first show that $q(J) = J/I$ is indeed an ideal of R/I, for J an ideal of R containing I. Let $x + I \in J/I$ and $r + I \in R/I$. Then, $(r + I)(x + I) = (r \cdot x) + I$. Since $x \in J$ and J is an ideal of R, we have $r \cdot x \in J$, so $(r + I)(x + I) \in J/I$. Also note that for all $x + I, y + I \in J/I$, we have $(x + I) + (y + I) = (x + y) + I \in J/I$ since $x, y \in J$ and J is an ideal of R. Thus, J/I is an ideal of R/I.

On the other hand, we show that $q^{-1}(K)$ is an ideal of R for K an ideal of R/I. Let $x, y \in q^{-1}(K)$. Then, $q(x), q(y) \in K$, so $q(x + y) = q(x) + q(y) \in K$ since K is an ideal of R/I. Thus, $x + y \in q^{-1}(K)$. Also, for any $r \in R$ and $x \in q^{-1}(K)$, we have $q(r), q(x) \in R/I$ and $q(x) \in K$, so $q(r \cdot x) = q(r) \cdot q(x) \in K$ since K is an ideal of R/I. Thus, $r \cdot x \in q^{-1}(K)$. Therefore, $q^{-1}(K)$ is an ideal of R. Also, if $x \in I$, then $q(x) = x + I = I$, which is the additive identity in R/I and thus belongs to every ideal of R/I. Therefore, $I \subseteq q^{-1}(K)$.

To show that the maps are inverses of each other is left as an exercise. ■

Theorem 1.33 (The *second isomorphism theorem* for rings)**.** *Let R be a ring, and let $S \subseteq R$ be a subring containing the unity. Let I be an ideal of R. Then, $S + I = \{s + i : s \in S, i \in I\}$ is a subring of R containing the unity, $S \cap I$ is an ideal of S, and there is a ring isomorphism*

$$(S + I)/I \cong S/(S \cap I). \tag{1.28}$$

Proof. Let $\alpha, \beta \in S + I$. Then $\alpha = s + x$ and $\beta = s' + y$ for some $s, s' \in S$ and $x, y \in I$. Thus, $\alpha + \beta = (s + s') + (x + y) \in S + I$ since S is a subring and I is an ideal. Also, $\alpha \cdot \beta = (s + x)(s' + y) = ss' + sy + xs' + xy \in S + I$ since $ss' \in S$, $sy, xs', xy \in I$. Therefore, $S + I$ is a subring of R containing the unity.

Note that the inclusion map $i : S \to R$ is a ring homomorphism. Thus, by the proposition above, $S \cap I = i^{-1}(I)$ is an ideal of S. Also, $I \subseteq S + I$ is an ideal of $S + I$. Now let $\varphi : S \to (S + I)/I$ be the map $\varphi = q \circ i$, where $q : S + I \to (S + I)/I$ is the quotient map. It is easy to verify that φ is a ring homomorphism with kernel

$$\ker \varphi = \{a \in S \mid q \circ i(a) = I\} = \{a \in S \mid a + I = I\} = S \cap I. \tag{1.29}$$

Moreover, φ is surjective since for any $s + i + I \in (S + I)/I$ where $s \in S$ and $i \in I$, we have $\varphi(s) = s + I = s + i + I$. Thus, by the first isomorphism theorem, we have the desired isomorphism. ■

January 30th.

Theorem 1.34 (The *third isomorphism theorem* for rings)**.** *Let R be a ring, and let $J \subseteq I$ be two ideals of R. Then, $I/J = \{a + J : a \in I\}$ is an ideal of the quotient ring R/J, and there is a ring isomorphism*

$$(R/J)/(I/J) \cong R/I. \tag{1.30}$$

Proof. Let $q_R : R \to R/J$ be the quotient map, and $q_{R/J} : R/J \to (R/J)/(I/J)$ be the quotient map. Thus, the composition $\varphi = q_{R/J} \circ q_R : R \to (R/J)/(I/J)$ is a surjective ring homomorphism. The kernel of φ is given by

$$\ker \varphi = \{x \in R \mid \varphi(x) = J + I/J\} = \{x \in R \mid q_R(x) \in I/J\} = \{x \in R \mid x + J \in I/J\} = I. \tag{1.31}$$

Thus, by the first isomorphism theorem, we have the desired isomorphism. ■

We look at some applications of the isomorphism theorems.

Example 1.35. Let $I = (5) \subseteq \mathbb{Z}[x]$. We claim that $\mathbb{Z}[x]/5\mathbb{Z}[x] \cong (\mathbb{Z}/5\mathbb{Z})[x]$. To see this, we make use of the first isomorphism theorem. Let $\varphi : \mathbb{Z}[x] \to (\mathbb{Z}/5\mathbb{Z})[x]$ be the map defined by

$$\varphi\left(a_0 + a_1x + a_2x^2 + \cdots + a_nx^n\right) = \bar{a}_0 + \bar{a}_1x + \bar{a}_2x^2 + \cdots + \bar{a}_nx^n, \tag{1.32}$$

where $\bar{a}_i$ is the image of a_i in $\mathbb{Z}/5\mathbb{Z}$ for all $0 \leq i \leq n$. It is easy to verify that φ is a surjective ring homomorphism with kernel $5\mathbb{Z}[x]$. Thus, by the first isomorphism theorem, we have the desired isomorphism.

Example 1.36. Let $(x) \subseteq \mathbb{Z}[x]$. We claim that $\mathbb{Z}[x]/(x) \cong \mathbb{Z}$. To see this, we make use of the second isomorphism theorem. Let $S = \mathbb{Z} \subseteq \mathbb{Z}[x]$. Then, $S + (x) = \mathbb{Z}[x]$ since for any $f(x) \in \mathbb{Z}[x]$, we have $f(x) = f(0) + (f(x) - f(0)) \in S + (x)$. Also, $S \cap (x) = \{0\}$ since the only constant polynomial in (x) is the zero polynomial. Thus, by the second isomorphism theorem, we have

$$\mathbb{Z}[x]/(x) \cong S/(S \cap (x)) = S/\{0\} \cong \mathbb{Z}. \tag{1.33}$$

Example 1.37. Again, let $I = (x^2 - 4, 2) \subseteq \mathbb{Z}[x]$. We claim the isomorphism

$$\mathbb{Z}[x]/(x^2 - 4, 2) \cong \mathbb{Z}/2\mathbb{Z}[x]/(x^2). \tag{1.34}$$

To see this, we make use of the third isomorphism theorem. Let $J = (2) \subseteq I$. Then, by the third isomorphism theorem, we have

$$\mathbb{Z}[x]/I \cong (\mathbb{Z}[x]/J)/(I/J) \cong (\mathbb{Z}/2\mathbb{Z})[x]/(x^2 - 4 + J) = (\mathbb{Z}/2\mathbb{Z})[x]/(x^2), \tag{1.35}$$

since $x^2 - 4 + J = x^2 + J$ in $(\mathbb{Z}/2\mathbb{Z})[x]$.

1.5 Prime and Maximal Ideals

Definition 1.38. Let R be a ring. An ideal $P \subseteq R$ is called a *prime ideal* if $P \neq R$ and for all $a, b \in R$ such that $a \cdot b \in P$, we have either $a \in P$ or $b \in P$.

Of course, the most common example of a prime ideal is (0_R) in an integral domain R. Another example is $(p) = p\mathbb{Z}$ in $\mathbb{Z}$ for any prime p. Note that if R is a field, then the only prime ideal of R is (0_R).

Theorem 1.39. *Let I be an ideal of a ring R. Then, I is a prime ideal if and only if the quotient ring R/I is an integral domain.*

Proof. Suppose that R/I is an integral domain. Let $a, b \in R$ such that $a \cdot b \in I$. Then,

$$(a + I)(b + I) = (a \cdot b) + I = I, \tag{1.36}$$

which is the zero element in R/I. Since R/I is an integral domain, either $a + I = I$ or $b + I = I$, which implies that either $a \in I$ or $b \in I$. Thus, I is a prime ideal. If we now suppose that I is a prime ideal, let $a + I, b + I \in R/I$ such that $(a + I)(b + I) = (a \cdot b) + I = I$. This implies that $a \cdot b \in I$, so either $a \in I$ or $b \in I$. Thus, either $a + I = I$ or $b + I = I$, so R/I is an integral domain. ■

One can also show that there is the natural bijection between ideals of R/I and ideals of R containing I restricts to a bijection between prime ideals of R/I and prime ideals of R containing I.

Example 1.40. We can use this theorem to show (x^2+1) is a prime ideal of $\mathbb{Z}[x]$. Indeed, look at the ring homomorphism $\varphi : \mathbb{Z}[x] \to \mathbb{Z}[i]$ defined by $\varphi(f) = f(i)$ for all $f \in \mathbb{Z}[x]$. For the kernel, let $f \in \ker\varphi$. By the division algorithm, there exist unique $q, r \in \mathbb{Z}[x]$ such that

$$f(x) = (x^2+1)q(x) + r(x), \tag{1.37}$$

where either $r(x) = 0$ or $\deg r < 2$. Plugging in $x = i$ gives $f(i) = 0 = r(i)$. The only way an at most linear polynomial $r(x)$ can be 0 at $x = i$ is if r is the zero polynomial. Hence, $\ker\varphi = (x^2+1)$. Since $\mathbb{Z}[i]$ is an integral domain, by the first isomorphism theorem, we have

$$\mathbb{Z}[x]/(x^2+1) \cong \mathbb{Z}[i] \tag{1.38}$$

showing that (x^2+1) is a prime ideal of $\mathbb{Z}[x]$.

Another notion is the maximal ideal.

Definition 1.41. Let R be a ring. An ideal $M \subseteq R$ is called a *maximal ideal* if $M \neq R$ and there are no ideals I of R such that $M \subsetneq I \subsetneq R$.

That is, if J is an ideal such that $M \subseteq J$, then either $J = M$ or $J = R$. For example, in $\mathbb{Z}$, the ideals $(p) = p\mathbb{Z}$ for prime p are maximal ideals. Note that if R is a field, then the only maximal ideal of R is (0_R). In fact, it may be shown that R is a field if and only if (0) and R are the only ideals.

February 2nd.

Theorem 1.42. *Let I be an ideal of a ring R. Then, I is a maximal ideal if and only if the quotient ring R/I is a field.*

Proof. We work with a set of equivalences. $I \subseteq R$ is a maximal ideal $\iff \{J \mid I \subseteq J \subseteq R\} = \{I, R\}$ $\iff \{K \mid K \text{ is an ideal of } R/I\} = \{I/I, R/I\} \iff R/I$ is a field. ■

A neat corollary of this theorem is that every maximal ideal is a prime ideal. Indeed, if M is a maximal ideal of R, then R/M is a field, and thus an integral domain. Therefore, by the previous theorem, M is a prime ideal. Another theorem guarantees the existence of maximal ideals.

Theorem 1.43. *Every non-zero ring has at least a maximal ideal.*

This theorem, though true in many common cases, requires Zorn's lemma for a general proof.

Proof. Let $(\Omega, \subseteq)$ be the set of all proper ideals of a ring R, ordered by inclusion. Note that $R \supsetneq (0) \in \Omega$, so $\Omega \neq \emptyset$. Zorn's lemma states that if every chain in Ω has an upper bound in Ω, then Ω has a maximal element. Let $\mathcal{C}$ be a chain in Ω. We claim that $I_{\mathcal{C}} = \bigcup_{I \in \mathcal{C}} I$ is an upper bound of $\mathcal{C}$ in Ω. Certainly, for any $I \in \mathcal{C}$, we have $I \subseteq J$. Also, we must verify that $I_{\mathcal{C}}$ is a proper ideal of R.

Let $x, y \in I_{\mathcal{C}}$, and $r \in R$. Then there exist two ideals $I, J \in \mathcal{C}$ such that $x \in I$ and $y \in J$. Since $\mathcal{C}$ is a chain, without loss of generality, suppose that $I \subseteq J$. Thus, $x, y \in J$, so $x + y \in R$ and $r \cdot x \in R$ since J is an ideal of R. Therefore, $I_{\mathcal{C}}$ is an ideal of R. Also, if $I_{\mathcal{C}} = R$, then $1_R \in I_{\mathcal{C}}$, so there exists some ideal $I \in \mathcal{C}$ such that $1_R \in I$. This implies that $I = R$, which contradicts the fact that I is a proper ideal. Thus, $I_{\mathcal{C}}$ is a proper ideal of R, so $I_{\mathcal{C}} \in \Omega$. Therefore, by Zorn's lemma, Ω has a maximal element, which is a maximal ideal of R. ■

Example 1.44. In $\mathbb{C}[x]$, the ideal (x) is maximal since $\mathbb{C}[x]/(x) \cong \mathbb{C}$, which is a field. However, (x^2) is not maximal since $(x^2) \subsetneq (x)$. Another reasoning is that $\mathbb{C}[x]/(x^2)$ is not a field since $(x + (x^2)) \cdot (x + (x^2)) = x^2 + (x^2) = 0 + (x^2)$; it is not an integral domain either, so (x^2) is not even prime. In fact, the maximal ideals of $\mathbb{C}[x]$ are precisely of the form $(x - a)$ for some $a \in \mathbb{C}$.

1.5.1 Jacobson Radical and Nilradical

February 5th.

Definition 1.45. The *Jacobson radical* of a non-zero ring R is defined as

$$\mathrm{Jac}(R) = \bigcap_{\substack{\mathfrak{m} \subseteq R \\ \mathfrak{m} \text{ is a maximal ideal}}} \mathfrak{m}. \tag{1.39}$$

It is the interseciton of ideals, so it is also an ideal. The *nilradical* of R is defined as

$$\mathrm{nil}(R) = \{x \in R \mid x^n = 0 \text{ for some } n \in \mathbb{N}\}. \tag{1.40}$$

The following are some properties of the Jacobson radical and nilradical.

Proposition 1.46. *Let R be a ring. Then,*

1. $\mathrm{nil}(R)$ *is an ideal of* R.
2. $\mathrm{nil}(R) \subseteq \mathrm{Jac}(R)$.
3. $x \in \mathrm{Jac}(R)$ *if and only if* $1 + ax$ *is a unit for all* $a \in R$.
4. $\mathrm{nil}(R) = \bigcap_{\substack{P \subseteq R \\ P \text{ is a prime ideal}}} P$.

Proof. 1. Let $x, y \in \mathrm{nil}(R)$ and $r \in R$. Then, there exist $m, n \in \mathbb{N}$ such that $x^m = 0$ and $y^n = 0$. Thus, $(x+y)^{m+n} = 0$ by the binomial theorem, so $x + y \in \mathrm{nil}(R)$. Also, $(r \cdot x)^m = r^m \cdot x^m = 0$, so $r \cdot x \in \mathrm{nil}(R)$. Therefore, $\mathrm{nil}(R)$ is an ideal of R.

2. Let $x \in \mathrm{nil}(R)$. Then, there exists some $n \in \mathbb{N}$ such that $x^n = 0$. Thus x^n belongs to every ideal of R, so in particular, $x^n \in P$ for every prime ideal and $x^n \in \mathfrak{m}$ for every maximal ideal. However, $x^n \in P$ implies that $x \in P$ since P is prime, and $x^n \in \mathfrak{m}$ implies that $x \in \mathfrak{m}$ since $\mathfrak{m}$ is prime as well. Therefore, x belongs to every prime ideal and every maximal ideal, so $x \in \mathrm{Jac}(R)$. Hence, $\mathrm{nil}(R) \subseteq \mathrm{Jac}(R)$.

3. For the forward implication, let $x \in \mathrm{Jac}(R)$ and $a \in R$. Note that ax cannot be a unit since it is contained in the (maximal) ideals $\mathfrak{m}$. Suppose $1 + ax$ is *not* a unit. Then $I = (1 + ax)$ is a proper ideal of R. If $q : R \to R/I$ is the quotient map, and $\bar{\mathfrak{m}}$ is a maximal ideal of R/I, then we claim that $\mathfrak{m} = q^{-1}(\bar{\mathfrak{m}})$ is a maximal ideal of R containing I. Certainly, $\mathfrak{m}$ is an ideal of R containing I by the proposition above. Also, if there exists some ideal J of R such that $\mathfrak{m} \subsetneq J \subsetneq R$, then $I \subseteq J \subsetneq R$, so $q(J)$ is an ideal of R/I such that $\bar{\mathfrak{m}} \subsetneq q(J) \subsetneq R/I$, contradicting the maximality of $\bar{\mathfrak{m}}$. Thus, $\mathfrak{m}$ is a maximal ideal of R. However, since $x \in \mathrm{Jac}(R)$, we have $x \in \mathfrak{m}$, so $ax \in I \subseteq \mathfrak{m}$. This implies that $1 = (1 + ax) - ax \in \mathfrak{m}$, a contradiction. Therefore, $1 + ax$ is a unit for all $a \in R$.

   For the converse, suppose $1 + ax$ is a unit for all $a \in R$. Let $\mathfrak{m}$ be any maximal ideal of R. If $x \notin \mathfrak{m}$, then the ideal $(\mathfrak{m}, x)$ properly contains $\mathfrak{m}$, so $(\mathfrak{m}, x) = R$. Thus, there exist $m \in \mathfrak{m}$ and $r \in R$ such that $1 = m + r \cdot x$. This implies that $1 - r \cdot x = m \in \mathfrak{m}$, contradicting the fact that $1 - r \cdot x$ is a unit. Therefore, $x \in \mathfrak{m}$. Since $\mathfrak{m}$ was an arbitrary maximal ideal, we have $x \in \mathrm{Jac}(R)$.

4. It is clear that $\mathrm{nil}(R) \subseteq \bigcap_{P \text{ prime}} P$ since maximal ideals are prime ideals. For the converse inclusion, let $x \in \bigcap_{P \text{ prime}} P$. We claim that x is nilpotent. Suppose not. Then, the set $S = \{x^n : n \in \mathbb{N}\}$ does not contain 0. Define the set $\Omega = \{I \subseteq R \mid I \text{ is an ideal and } I \cap S = \emptyset\}$. Inclusion $\subseteq$ is a partial order on Ω. Let $\mathcal{C}$ be a chain in Ω and let $I = \bigcup_{J \in \mathcal{C}} J$. We claim that I is an upper bound of $\mathcal{C}$ in Ω. It is an ideal since if $z, y \in I$, then $z \in J_1$ and $y \in J_2$ with $J_1 \subseteq J_2$. Thus, $z, y \in J_1 \cup J_2$, so $z + y \in J_1 \cup J_2 \subseteq I$. Also, for any $r \in R$ and $z \in I$, we have $z \in J$ for some $J \in \mathcal{C}$, so $r \cdot z \in J \subseteq I$. Therefore, I is an ideal. Moreover, if $J \cap S = \emptyset$ for all $J \in \mathcal{C}$ tells us $I \cap S = \emptyset$. Hence, I is a valid upper bound, and Zorn's lemma guarantess the existence of a maximal P in Ω. We further claim that P is a prime ideal of R. Indeed, $P \subsetneq R$ since $P \cap S = \emptyset$. Let $uv \in P$ for some $u, v \in R$. If both $u, v \notin P$, then the ideals (P, u) and (P, v) properly contain P; that is, $au + y = x^n$ and $bv + z = x^m$ for some $a, b \in R$, $y, z \in P$, and $n, m \in \mathbb{N}$. Thus, $(au + y)(bv + z) = x^{n+m} \in S$, but

$(au+y)(bv+z) = abuv + az + by + yz \in P$ since $uv \in P$ and $y, z \in P$. This contradicts the fact that $P \cap S = \emptyset$. Therefore, either $u \in P$ or $v \in P$, so P is a prime ideal. However, by construction, x is in every prime ideal of R, so $x \in P$, contradicting the fact that $P \cap S = \emptyset$. Hence, x is nilpotent, so $x \in \mathrm{nil}(R)$.

■

February 6th.

Let us look at some examples.

Example 1.47. In $\mathbb{Z}$, $\mathrm{nil}(\mathbb{Z}) = \{0\}$ since the only nilpotent element is 0. Also, $\mathrm{Jac}(\mathbb{Z}) = \{0\}$ since the intersection of all maximal ideals (p) for prime p is $\{0\}$. In the ring $\mathbb{Z}/6\mathbb{Z}$, the nilradical is $\{0\}$ even though it is not an integral domain. The Jacobson radical is $\{0\}$ as well since the maximal ideals are (2) and (3), whose intersection is $\{0\}$.

Example 1.48. For a non-trivial example, let us look at the ring $R = \mathbb{Z}/4\mathbb{Z}$. The nilradical of R is $\{0, 2\}$ since $2^2 = 0$ in R. The Jacobson radical of R is also $\{0, 2\}$ since the only maximal ideal of R is (2). If we take $R = \mathbb{Q}[x, y]/(x^2, y)$, then $\mathrm{nil}(R) = (x)$ since x is nilpotent. The Jacobson radical of R is also (x).

Example 1.49. For an example where the Jacobson radical is strictly larger than the nilradical, consider the power series ring $R = \mathbb{Q}[[x]]$. If $f \in \mathbb{Q}[[x]]$, then f is a unit if and only if the constant term of f is non-zero; if $f = a_0(1-g)$ where $a_0 \in \mathbb{Q}$ and $g \in (x)$, then $f^{-1} = a_0^{-1}(1-g)^{-1}$, where $(1-g)^{-1}$ is given by the geometric series expansion $(1-g)^{-1} = 1 + g + g^2 + \cdots$. Thus, $\mathrm{Jac}(R) = (x)$ since $1 + ax$ is a unit for all $a \in R$. However, $\mathrm{nil}(R) = \{0\}$ since there are no non-zero nilpotent elements in R.

1.6 Product of Rings

Let $R_1, R_2, \ldots, R_n$ be rings. We can define the *product ring* $R_1 \times R_2 \times \cdots \times R_n$ as the set of all n-tuples $(r_1, r_2, \ldots, r_n)$ where $r_i \in R_i$ for all $1 \leq i \leq n$, with addition and multiplication defined componentwise. It is easy to verify that $R_1 \times R_2 \times \cdots \times R_n$ is also a ring, and the unity is given by $(1_{R_1}, 1_{R_2}, \ldots, 1_{R_n})$. The *projection map* $p_i : R_1 \times R_2 \times \cdots \times R_n \to R_i$ is defined by $p_i(r_1, r_2, \ldots, r_n) = r_i$ for all $1 \leq i \leq n$. It is, again, easy to verify that p_i is a surjective ring homomorphism with kernel $R_1 \times \cdots \times R_{i-1} \times \{0_{R_i}\} \times R_{i+1} \times \cdots \times R_n$. Conversely, the *inclusion map* $e_i : R_i \to R_1 \times R_2 \times \cdots \times R_n$ is defined by $e_i(r) = (0_{R_1}, \ldots, 0_{R_{i-1}}, r, 0_{R_{i+1}}, \ldots, 0_{R_n})$ for all $r \in R_i$. It is easy to verify that e_i is an injective ring homomorphism with image $\{0_{R_1}\} \times \cdots \times \{0_{R_{i-1}}\} \times R_i \times \{0_{R_{i+1}}\} \times \cdots \times \{0_{R_n}\}$.

Regarding ideals, we have the following proposition.

Proposition 1.50. *Let $R_1, R_2, \ldots, R_n$ be rings. Then, every ideal of the product ring $R_1 \times R_2 \times \cdots \times R_n$ is of the form $I_1 \times I_2 \times \cdots \times I_n$ where I_i is an ideal of R_i for all $1 \leq i \leq n$.*

Proof. Let I be an ideal of $R_1 \times R_2 \times \cdots \times R_n$. For each $1 \leq i \leq n$, let $I_i = p_i(I)$. We claim that each I_i is an ideal of R_i and $I = I_1 \times I_2 \times \cdots \times I_n$. Let $x_i, y_i \in I_i$ and $r_i \in R_i$. Then, there exist $x = (x_1, \ldots, x_n), y = (y_1, \ldots, y_n) \in I$ such that $p_i(x) = x_i$ and $p_i(y) = y_i$. Thus, $x + y \in I$ since I is an ideal of the product ring, so $x_i + y_i = p_i(x + y) \in I_i$. Also, $r_i \cdot x_i = p_i(e_i(r_i) \cdot x) \in I_i$ since $e_i(r_i) \cdot x \in I$. Therefore, I_i is an ideal of R_i.

Let use first show that $I \supseteq I_1 \times I_2 \times \cdots \times I_n$. Let $(a_1, a_2, \ldots, a_n) \in I_1 \times I_2 \times \cdots \times I_n$. Let $x^{(i)} \in I$ such that $p_i(x^{(i)}) = a_i$ for all $1 \leq i \leq n$. Then $e_i(1_{R_i}) \cdot x^{(i)} \in I$ since I is an ideal of the product ring, and so $\sum_{i=1}^n e_i(1_{R_i}) \cdot x^{(i)} \in I$ as well. However, $\sum_{i=1}^n e_i(1_{R_i}) \cdot x^{(i)} = (a_1, a_2, \ldots, a_n)$, so $(a_1, a_2, \ldots, a_n) \in I$. Therefore, $I \supseteq I_1 \times I_2 \times \cdots \times I_n$. For the converse inclusion $I \subseteq I_1 \times I_2 \times \cdots \times I_n$, let $(a_1, a_2, \ldots, a_n) \in I$. Then, $a_i = p_i(a_1, a_2, \ldots, a_n) \in I_i$ for all $1 \leq i \leq n$. Thus, $(a_1, a_2, \ldots, a_n) \in I_1 \times I_2 \times \cdots \times I_n$. Therefore, $I = I_1 \times I_2 \times \cdots \times I_n$. ■

We now wish to study the prime ideals of the product ring.

Proposition 1.51. *Let $R_1, R_2, \ldots, R_n$ be rings. Then $I \subseteq R_1 \times R_2 \times \cdots \times R_n$ is a prime ideal if and only if $p_i(I) = R_i$ for all $1 \leq i \leq n$ except for one index $1 \leq k \leq n$ such that $p_k(I)$ is a prime ideal of R_k.*

Proof. For the forward direction, let $I = I_1 \times \cdots I_n$ (by the previous proposition), where $I_i = p_i(I)$ is an ideal of R_i. Note that I is prime if and only if $(R_1 \times \cdots \times R_n)/I \cong R_1/I_1 \times \cdots R_n/I_n$ is an integral domain. This implies that R_i/I_i is an integral domain for one index $1 \leq k \leq n$ and $R_i/I_i \cong \{0\}$ for all $i \neq k$. It it were otherwise, we could find two indices $i \neq j$ such that R_i/I_i and R_j/I_j are both non-trivial, and $e_i(1_{R_i}) + I$ and $e_j(1_{R_j}) + I$ are two non-zero elements in $R_1/I_1 \times \cdots R_n/I_n$ whose product is zero, contradicting the fact that $R_1/I_1 \times \cdots R_n/I_n$ is an integral domain. Thus, $p_i(I) = I_i = R_i$ for all $i \neq k$, and $p_k(I) = I_k$ is a prime ideal of R_k.

Conversely, suppose that $p_i(I) = R_i$ for all $1 \leq i \leq n$ except for one index $1 \leq k \leq n$ such that $p_k(I)$ is a prime ideal of R_k. Then, $(R_1 \times \cdots \times R_n)/I \cong R_1/R_1 \times \cdots R_k/p_k(I) \times \cdots R_n/R_n \cong R_k/p_k(I)$ is an integral domain since $p_k(I)$ is a prime ideal of R_k. Thus, I is a prime ideal of the product ring. ■

1.6.1 Idempotents

Definition 1.52. Let R be a ring. An element $x \in R$ is called an *idempotent* if $x^2 = x$.

Example 1.53. Trivially, every ring has two idempotents, namely 0_R and 1_R. If R_1 and R_2 are rings, then $(1,1), (0,0), (1,0), (0,1)$ are idempotents of the product ring $R_1 \times R_2$.

Proposition 1.54. *Let R be a ring and $e \in R$ be an idempotent. Then $1 - e$ is also an idempotent, and $R \cong eR \times (1-e)R$.*

Proof. Note that $(1-e)e = e - e^2 = 0$, and $(1-e)^2 = (1-e)(1-e) = 1-e$. To show the isomorphism, let $\varphi : R \to eR \times (1-e)R$ be defined by $\varphi(r) = (er, (1-e)r)$ for all $r \in R$. It is easy to verify that φ is a ring homomorphism. If we define $\psi : eR \times (1-e)R \to R$ by $\psi(x,y) = x + y$ for all $x \in eR$ and $y \in (1-e)R$, then ψ is also a ring homomorphism. Moreover, $\psi \circ \varphi(r) = \psi(er, (1-e)r) = er + (1-e)r = r$ for all $r \in R$, and $\varphi \circ \psi(x,y) = \varphi(x+y) = (e(x+y), (1-e)(x+y)) = (x,y)$ for all $x \in eR$ and $y \in (1-e)R$. Thus, φ is an isomorphism, so $R \cong eR \times (1-e)R$. ■

February 9th.

An interesting consequence is the chinese remainder theorem, which is abstracted from the original number-theoretic version. For the below theorem, we define the product of two ideals I and J of a ring R as

$$IJ = \{a_1 b_1 + \cdots + a_n b_n \mid a_i \in I, b_i \in J, n \in \mathbb{N}\}. \tag{1.41}$$

One may verify that IJ is an ideal of R. Moreover, $IJ \subseteq I \cap J$ since $a_i b_i \in I$ and $a_i b_i \in J$ for all $1 \leq i \leq n$. However, it is not necessarily the case that $IJ = I \cap J$. For example, if $R = \mathbb{Z}$ and $I = (2)$ and $J = (4)$, then $IJ = (8)$ but $I \cap J = (4)$.

Theorem 1.55 (The *chinese remainder theorem*)**.** *Let R be a ring, and $I_1, I_2, \ldots, I_k$ be ideals of R which are pairwise co-maximal; that is, for all $1 \leq i \neq j \leq k$, we have $I_i + I_j = R$. Then $I_1 \cap I_2 \cap \cdots \cap I_k = I_1 I_2 \cdots I_k$. Moreover, the natural homomorphism $\varphi : R \to R/I_1 \times R/I_2 \times \cdots \times R/I_k$ defined by $\varphi(r) = (r + I_1, r + I_2, \ldots, r + I_k)$ is surjective with kernel $I_1 I_2 \cdots I_k$, so*

$$R/(I_1 I_2 \cdots I_k) \cong R/I_1 \times R/I_2 \times \cdots \times R/I_k. \tag{1.42}$$

We first determine how this abstract version of the chinese remainder theorem implies the original number-theoretic version. Let $n_1, n_2, \ldots, n_k$ be pairwise coprime positive integers, and let $m = n_1 n_2 \cdots n_k$. Then, the ideals $(n_1), (n_2), \ldots, (n_k)$ of $\mathbb{Z}$ are pairwise co-maximal since $(n_i) + (n_j) = (1)$ for all $1 \leq i \neq j \leq k$. Thus, by the chinese remainder theorem, we have

$$\mathbb{Z} \to \mathbb{Z}/n_1\mathbb{Z} \times \mathbb{Z}/n_2\mathbb{Z} \times \cdots \times \mathbb{Z}/n_k\mathbb{Z}, \quad x \mapsto (x + n_1\mathbb{Z}, x + n_2\mathbb{Z}, \ldots, x + n_k\mathbb{Z}) \tag{1.43}$$

is surjective with kernel (m). That is, given any $(a_1 + n_1\mathbb{Z}, a_2 + n_2\mathbb{Z}, \ldots, a_k + n_k\mathbb{Z}) \in \mathbb{Z}/n_1\mathbb{Z} \times \mathbb{Z}/n_2\mathbb{Z} \times \cdots \times \mathbb{Z}/n_k\mathbb{Z}$, there exists some integer x such that $x \equiv a_i \pmod{n_i}$ for all $1 \leq i \leq k$, and any two such integers are congruent modulo m. Moreover, we have the isomorphism

$$\mathbb{Z}/m\mathbb{Z} \cong \mathbb{Z}/n_1\mathbb{Z} \times \mathbb{Z}/n_2\mathbb{Z} \times \cdots \times \mathbb{Z}/n_k\mathbb{Z}. \tag{1.44}$$

Proof. We first show for $k = 2$. Let I_1 and I_2 be two ideals of R such that $I_1 + I_2 = R$. Then, there exist $a \in I_1$ and $b \in I_2$ such that $a + b = 1$. We already know that $I_1I_2 \subseteq I_1 \cap I_2$, so we only need to show the converse inclusion. Let $x \in I_1 \cap I_2$. Then, $x = x(a + b) = xa + xb \in I_1I_2$ since $xa \in I_1I_2$ and $xb \in I_1I_2$. Therefore, $I_1 \cap I_2 = I_1I_2$. We now wish to show the natural map $R \to R/I_1 \times R/I_2$ defined by $r \mapsto (r + I_1, r + I_2)$ is surjective with kernel I_1I_2. It is easy to verify that the map is a ring homomorphism. For surjectivity, since any element of $R/I_1 \times R/I_2$ is of the form $(r_1 + I_1, r_2 + I_2)$ for some $r_1, r_2 \in R$, we only show that $(1 + I_1, 0 + I_2)$ and $(0 + I_1, 1 + I_2)$ are in the image of the map. Since $a \in I_1$, we have $a + I_1 = 0 + I_1$, so $\varphi(b) = (b + I_1, b + I_2) = (1 + I_1, 0 + I_2)$. Similarly, since $b \in I_2$, we have $b + I_2 = 0 + I_2$, so $\varphi(a) = (a + I_1, a + I_2) = (0 + I_1, 1 + I_2)$. Therefore, the map is surjective. For the kernel, if $r \in R$ is such that $\varphi(r) = (r + I_1, r + I_2) = (0 + I_1, 0 + I_2)$, then $r \in I_1 \cap I_2 = I_1I_2$. Conversely, if $r \in I_1I_2$, then $r \in I_1 \cap I_2$, so $\varphi(r) = (0 + I_1, 0 + I_2)$. Thus, the kernel of the map is I_1I_2. For arbitrary k, we claim that I_1 and $I_2 \cdots I_k$ are co-maximal ideals. Since I_1 and I_j are co-maximal for all $2 \leq j \leq k$, there exist $x_j \in I_1$ and $y_j \in I_j$ such that $x_j + y_j = 1$ for all $2 \leq j \leq k$. Then

$$(x_2 + y_2)(x_3 + y_3) \cdots (x_k + y_k) = 1 \implies \alpha + y_2y_3 \cdots y_k = 1 \tag{1.45}$$

where $\alpha \in I_1$ since $x_j \in I_1$ for all $2 \leq j \leq k$, and $y_2 \cdots y_k \in I_2 \cdots I_k$. Thus, I_1 and $I_2 \cdots I_k$ are co-maximal. Now induction can be applied on k to show $I_1 \cap I_2 \cap \cdots \cap I_k = I_1 \cap I_2 \cdots I_k = I_1I_2 \cdots I_k$. To show surjectivity of the natural map $R \to R/I_1 \times R/I_2 \times \cdots \times R/I_k$, we show $e_j := (0 + I_1, \ldots, 0 + I_{j-1}, 1 + I_j, 0 + I_{j+1}, \ldots, 0 + I_k)$ is in the image of the map for all $1 \leq j \leq k$. Since I_j and $I_1 \cdots I_{j-1}I_{j+1} \cdots I_k$ are co-maximal, there exist $x \in I_j$ and $y \in I_1 \cdots I_{j-1}I_{j+1} \cdots I_k$ such that $x + y = 1$. Then $\varphi(y) = (y + I_1, y + I_2, \ldots, y + I_k) = e_j$ since $y \in I_i$ for all $i \neq j$ and $y + I_j = 1 + I_j$. Therefore, the natural map is surjective with kernel $I_1I_2 \cdots I_k$, so we have the desired isomorphism $R/(I_1I_2 \cdots I_k) \cong R/I_1 \times R/I_2 \times \cdots \times R/I_k$. ■

1.7 Generalizing Properties of Integers

February 12th.

The ring of integers has several properties, which we axiomatize as properties for more general rings.

Definition 1.56. Let R be an integral domain. Let $N : R \setminus \{0\} \to \mathbb{Z}_{\geq 0}$ be a function such that for all $a, b \in R \setminus \{0\}$, there exists $q, r \in R$ such that $a = bq + r$ and either $r = 0$ or $N(r) < N(b)$. Then, we say that R is a *euclidean domain* with respect to the euclidean function N.

Of course, the ring of integers $\mathbb{Z}$ is a euclidean domain with respect to the euclidean function $N : \mathbb{Z} \setminus \{0\} \to \mathbb{Z}_{\geq 0}$ defined by $N(n) = |n|$ for all $n \in \mathbb{Z} \setminus \{0\}$. We know this to satisfy the condition that given any $a, b \in \mathbb{Z} \setminus \{0\}$, there exist $q, r \in \mathbb{Z}$ such that $a = bq + r$ and $0 \leq r < |b|$. If k denotes a field, then $k[x]$ is a euclidean domain with respect to the euclidean function $N : k[x] \setminus \{0\} \to \mathbb{Z}_{\geq 0}$ defined by $N(f) = \deg(f)$ for all $f \in k[x] \setminus \{0\}$. We know this to satisfy the condition that given any $f, g \in k[x] \setminus \{0\}$, there exist $q, r \in k[x]$ such that $f = gq + r$ and either $r = 0$ or $\deg(r) < \deg(g)$.

Example 1.57. A *valuation map* on a field k is a surjective map $v : k \to \mathbb{Z} \cup \{\infty\}$ such that $v(0) = \infty$, $v(ab) = v(a) + v(b)$ and $v(a + b) \geq \min\{v(a), v(b)\}$ for all $a, b \in k$. If v is a valuation map on k, then $R_v = \{x \in k : v(x) \geq 0\}$ is a valid ring with respect to the usual addition and multiplication on k termed the *valuation ring*. Moreover, R_v is a euclidean domain with respect to the euclidean function $N : R_v \setminus \{0\} \to \mathbb{Z}_{\geq 0}$ defined by $N(x) = v(x)$ for all $x \in R_v \setminus \{0\}$. To show that N satisfies the condition in the definition of euclidean domain, let $a, b \in R_v \setminus \{0\}$. If $v(a) < v(b)$, then we can take $q = 0$ and $r = a$. If $v(a) \geq v(b)$, then we can take $q = a/b$ so that $v(a/b) = v(a) - v(b) \geq 0$, so $q \in R_v$, and $r = a - bq = 0$. Therefore, R_v is a euclidean domain with respect to the euclidean function N.

Definition 1.58. Let R be an integral domain. We say that R is a *principal ideal domain* if every ideal of R is principal; that is, for every ideal I of R, there exists some $a \in R$ such that $I = (a)$.

In other words, every ideal of R is generated by a single element. The ring of integers $\mathbb{Z}$ is a principal ideal domain since every ideal of $\mathbb{Z}$ is of the form (n) for some $n \in \mathbb{Z}$. More trivially, $\mathbb{Q}$ and $\mathbb{R}$ are principal ideal domains since the only ideals of $\mathbb{Q}$ and $\mathbb{R}$ are the zero ideal and the whole ring. Generally, we can show the following proposition.

Proposition 1.59. *Let R be a euclidean domain. Then R is a principal ideal domain.*

Proof. Let $N : R \setminus \{0\} \to \mathbb{Z}_{\geq 0}$ be a euclidean function on R. Let I be an ideal of R. If $I = \{0\}$, then $I = (0)$ is principal. Else, let $x \in I$ be such that $N(x) = \min\{N(y) : y \in I \setminus \{0\}\}$. We claim that $I = (x)$. It is clear that $(x) \subseteq I$ since $x \in I$ and I is an ideal. For the converse inclusion, let $y \in I \setminus \{0\}$. Then, there exist $q, r \in R$ such that $y = xq + r$ and either $r = 0$ or $N(r) < N(x)$. If $r = 0$, then $y = xq \in (x)$. If $r \neq 0$, then $r = y - xq \in I$, contradicting the choice of x since $N(r) < N(x)$. Therefore, we must have $r = 0$, so $y \in (x)$. Thus, $I = (x)$ is principal. ■

Definition 1.60. Let R be a ring. A non-zero element $x \in R$ is called a *prime element* if the ideal (x) generated by x is a non-zero prime ideal of R. A non-zero element $x \in R$ is called an *irreducible element* if x is not a unit and whenever $x = ab$ for some $a, b \in R$, then either a or b is a unit.

That is, if $x \mid ab$, then $x \mid a$ or $x \mid b$ for all $a, b \in R$ (note that $x \mid a$ means there exists $c \in R$ such that $a = cx$) for a prime x. For example, in $\mathbb{Z}$, the prime elements and irreducible elements are exactly the prime numbers and their negatives. In $k[x]$ where k is a field, the prime elements and irreducible elements are exactly the irreducible polynomials.

Proposition 1.61. *Let R be an integral domain. Then every prime element of R is irreducible.*

Proof. Let $x = ab$ for $a, b \in R$. Then $x \mid ab$, so $x \mid a$ or $x \mid b$ since x is prime. If $x \mid a$, then there exists some $c \in R$ such that $a = cx$. Thus, $x = ab = cxb$, so $(1 - cb)x = 0$. Since R is an integral domain and $x \neq 0$, we must have $cb = 1$, so b is a unit. If $x \mid b$, then there exists some $d \in R$ such that $b = dx$. Thus, $x = ab = adx$, so $(1 - ad)x = 0$. Since R is an integral domain and $x \neq 0$, we must have $ad = 1$, so a is a unit. Therefore, either a or b is a unit, so x is irreducible. ■

Note that in a field, there are no prime elements. In a valuation ring, x is a prime element if and only if $v(x) = 1$.

Proposition 1.62. *Let R be a principal ideal domain. Then every irreducible element of R is prime.*

Proof. Consider $I = (x)$. Let $\mathfrak{m}$ be a maximal ideal of R containing I. Since R is a principal ideal domain, $\mathfrak{m} = (a)$ for some $a \in R$. Since $I \subseteq \mathfrak{m}$, we have $x \in (a)$, so there exists some $b \in R$ such that $x = ab$. Since x is irreducible, either a or b is a unit. If a is a unit, then $\mathfrak{m} = (a) = R$, contradicting the fact that $\mathfrak{m}$ is a maximal ideal. Thus, b is a unit and $a = b^{-1}x$. Thus $\mathfrak{m} = (a) = (b^{-1}x) \subseteq (x) = I$. Since $\mathfrak{m}$ is a maximal ideal containing I, we must have $\mathfrak{m} = I$. Therefore, I is a prime ideal, so x is a prime element. ■

Corollary 1.63. *In a principal ideal domain, every non-zero prime ideal is a maximal ideal.*

Definition 1.64. Let R be a ring, and $a, b \in R$. We say that $d \in R$ is a *greatest common divisor* of a and b, and we write $d = \gcd(a, b)$, if $d \mid a$ and $d \mid b$, and if there exists some $d' \in R$ such that $d' \mid a$ and $d' \mid b$, then $d' \mid d$.

Note that the gcd here may not be unique. For example, in $\mathbb{Z}$, the gcd of 6 and -9 is 3 and -3.

Proposition 1.65. *Let R be a ring, and $a, b \in R$ such that (a, b) is a principal ideal. Then the* gcd *of a and b exists, and if $(d) = (a, b)$, then d is a* gcd *of a and b. Moreover, $d = ax + by$ for some $x, y \in R$.*

Proof. If $(a,b) = (d)$, then $d \mid a$ and $d \mid b$. If $d' \in R$ is such that $d' \mid a$ and $d' \mid b$, then $(a,b) \subseteq (d')$, so $(d) \subseteq (d')$, so $d' \mid d$. Thus, d is a gcd of a and b. Since $(d) = (a,b)$, there exist $x, y \in R$ such that $d = ax + by$. ■

Of course, the notion of gcd exists for $\mathbb{Z}$. As another example, look at this ring $\mathbb{Q}[x,y]$. By inspection, 1 is a gcd of x and y, since $1 \mid x$ and $1 \mid y$, and if $d' \in \mathbb{Q}[x,y]$ is such that $d' \mid x$ and $d' \mid y$, then $d' \in \mathbb{Q}$, so $d' \mid 1$. In fact, any $a \in \mathbb{Q}^\times$ is a gcd of x and y.

February 13th.

Example 1.66. Consider the rings $\mathbb{Z}[i]$ and $\mathbb{Z}[\sqrt{-3}]$. The ring $\mathbb{Z}[i]$ is a euclidean domain with respect to the euclidean function $N : \mathbb{Z}[i] \setminus \{0\} \to \mathbb{Z}_{\geq 0}$ defined by $N(a+bi) = a^2 + b^2$ for all $a + bi \in \mathbb{Z}[i] \setminus \{0\}$. Indeed, given any $\alpha = a + ib, \beta = c + id \in \mathbb{Z}[i]$, for $a, b, c, d \in \mathbb{Z}$, we wish to find $\gamma, r \in \mathbb{Z}[i]$ such that $\alpha = \beta\gamma + r$ and either $r = 0$ or $N(r) < N(\beta)$. Consider

$$\frac{\alpha}{\beta} = \frac{(a+ib)(c-id)}{(c+id)(c-id)} = u + iv \tag{1.46}$$

where $u, v \in \mathbb{Q}$. So there exist $p, q \in \mathbb{Z}$ such that $|p - u| \leq 1/2$ and $|q - v| \leq 1/2$. Thus,

$$\alpha = \beta(u+iv) = \beta(p+iq) + \beta((u-p) + i(v-q)) = \beta(p+iq) + r \tag{1.47}$$

where $N(r) = N(\beta)N((u-p)+i(v-q)) \leq N(\beta)\frac{1}{2} < N(\beta)$. Thus, $\gamma = p+iq$ and $r = \beta((u-p)+i(v-q))$ satisfy the required condition. Therefore, $\mathbb{Z}[i]$ is a euclidean domain, so $\mathbb{Z}[i]$ is a principal ideal domain. Thus, $\mathbb{Z}[i]$ is a principal ideal domain, so every irreducible element of $\mathbb{Z}[i]$ is prime. In particular, $1+i$ is an irreducible element of $\mathbb{Z}[i]$, so $1+i$ is prime.
On the other hand, the ring $\mathbb{Z}[\sqrt{-3}]$ is not a principal ideal domain; consider $I = (2, 1+\sqrt{-3})$. We show that $I \cap \mathbb{Z} = (2)$. Since $2 \in I$, we have $(2) \subseteq I \cap \mathbb{Z}$. For the converse inclusion, let $x \in I \cap \mathbb{Z}$. Then, there exist $r, s \in \mathbb{Z}[\sqrt{-3}]$ such that $x = r(1+\sqrt{-3}) + s2$. Since x is also an integer, we can write $x = \alpha(1+\sqrt{-3}) + 2\beta$ where $\alpha \in \mathbb{Z}[-\sqrt{3}]$ and $\beta \in \mathbb{Z}$. But this means that $\alpha(1+\sqrt{-3})$ must be an integer, so taking norms gives $(x-2\beta)^2 = \alpha\bar{\alpha}4$, so $4 \mid (x-2\beta)^2$, so $2 \mid x - 2\beta$, so $2 \mid x$. Thus, $x \in (2)$, so $I \cap \mathbb{Z} = (2)$. Hence, $I \subsetneq \mathbb{Z}[\sqrt{-3}]$. If I were principal, then $I = (a + b\sqrt{-3})$ for some $a, b \in \mathbb{Z}$. If $b = 0$, then $I = (a)$ implies $I = 2\mathbb{Z}[\sqrt{-3}]$ which makes the contradiction $1 + \sqrt{-3} \notin 2\mathbb{Z}[\sqrt{-3}]$; b has to be non-zero. We must have, as $2 \in I$, $2 = (a + b\sqrt{-3})(c + d\sqrt{-3})$ for some $c, d \in \mathbb{Z}$. Taking norms gives $4 = (a^2 + 3b^2)(c^2 + 3d^2)$. Since $b \neq 0$, $b = \pm 1$ and $a = \pm 1$ must be the only possibilities. This again must mean that $c^2 + 3d^2 = 1$ which implies $c = \pm 1$ and $d = 0$. But this gives $2 = \pm(1 \pm \sqrt{-3})$, which is nonsense. Hence I cannot be principal.

A more general result states that if $D \equiv 1 \pmod 4$ and D is square-free, then $\mathbb{Z}[\sqrt{D}]$ cannot be a principal ideal domain.

Proposition 1.67. *Let R be a ring such that $R[x]$ is a principal ideal domain. Then R is a field.*

Proof. Since $R \subseteq R[x]$ is a subring, R is an integral domain. Let $\varphi : R[x] \to R$ be the evaluation homomorphism defined by $\varphi(f) = f(0)$ for all $f \in R[x]$. This map is surjective. Moreover, $\ker \varphi = (x) \subseteq R[x]$. So $R[x]/(x) \cong R$, implying that $R[x]/(x)$ is an integral domain and (x) is a prime ideal. We had shown that any non-zero prime ideal in a principal ideal domain is maximal, so (x) is maximal. Thus, $R[x]/(x) \cong R$ is a field, so R is a field. ■

Definition 1.68. For a ring R, the map $N : R \to \mathbb{Z}_{\geq 0}$ is called a *Dedekind-Hasse* norm if $N(0) = 0$, and for all $a, b \in R \setminus \{0\}$, either $b \mid a$ or there exists $0 \neq r \in (a,b)$ such that $N(r) < N(b)$.

Proposition 1.69. *If R is an integral domain that admits a Dedekind-Hasse norm, then R is a principal ideal domain. The converse also holds.*

Proof. Let $I \subseteq R$ be an ideal. Let $x \in I \setminus \{0\}$ be such that $N(x)$ is the least among $\{N(y) \mid y \in I, y \neq 0\}$. $(x) \subseteq I$ is clear. Let $y \in I \setminus \{0\}$. So either $x \mid y$ or there exists $0 \neq r \in (x, y)$ such that $N(r) < N(x)$. This, however, is a contradiction to the choice of x. Hence $x \mid y$ and $y \in (x)$. Thus, $I = (x)$ is principal. ■

Example 1.70. The ring $\mathbb{Z}[(1+\sqrt{-19})/2]$ is a principal ideal domain but not a euclidean domain. Note that $N : \mathbb{Z}[\omega = (1+\sqrt{-19})/2] \setminus \{0\} \to \mathbb{Z}_{\geq 0}$ defined by $a+b\omega \mapsto a^2+ab+5b^2$ is a Dedekind-Hasse norm.

Proposition 1.71. *Let R be a euclidean domain but not a field. Then it contains a* 'universal side divisor'*; that is, there exists a non-zero non-unit $u \in R$ such that for all non-zero $x \in R$, there exists $q \in R$ satisfying that $r = x - qu$ is either 0 or a unit*

Proof. Let $u \in \mathbb{R} \setminus \{0\}$ be such that $N(u)$ is least among all non-zero non-unit elements of R. We claim that u satisfies this property. Let $x \in R \setminus \{0\}$. Then there exist $q, r \in R$ such that $x = qu + r$ satisfying either $r = 0$ or $N(r) < N(u)$. Since u is a non-unit, $N(u) > 0$, so $N(r) < N(u)$ implies that r is a unit. Thus, $r = x - qu$ is either 0 or a unit, so the claim holds. ■

One may check that $\mathbb{Z}[(1+\sqrt{-19})/2]$ has no universal side division, and ± 1 are the only units in the ring.

Definition 1.72. Let R be an integral domain. We say that R is a *unique factorization domain* if for all non-zero non-unit $x \in R$, x can be written as $x = p_1 \cdots p_n$ for some $n \geq 1$ and p_i irreducibles, and if $x = p_1 \cdots p_n = q_1 \cdots q_m$ for some $m \geq 1$ and irreducibles q_j, then $n = m$ and, after reordering, $p_i = u_i q_i$ for some unit u_i for all $1 \leq i \leq n$.

For example, both $\mathbb{Z}$ and $k[x]$ where k is a field are UFDs. As a small definition, $x, y \in R$ are termed *associates* if $x = uy$ for some unit $u \in R$. Again, in $\mathbb{Z}$, $-n$ and n are associates, and in $\mathbb{Q}[x]$, f and cf for some $c \in \mathbb{Q}^\times$ are associates. Lastly, in $\mathbb{Z}[i]$, $n, -n, ni, -ni$ are all associates for any $n \in \mathbb{Z}$. Thus the above definition is saying that the factorization of x into irreducibles is unique up to reordering and up to associates. For non-examples, $\mathbb{Z}[\sqrt{5}]$ is an integral domain but not a unique factorization domain since $-4 = -2 \cdot 2 = (1+\sqrt{5})(1-\sqrt{5})$ are two distinct factorizations of -4 into irreducibles, and the irreducibles $2, 1+\sqrt{5}, 1-\sqrt{5}$ are not associates of each other.

February 16th.

Example 1.73. As a more concrete example of a non-UFD, consider the quotient ring $\bar{R} = \mathbb{Q}[x, y, z, w]/(xy - zw)$. Suppose $\bar{x} = \bar{f}\bar{g}$ for some $\bar{f}, \bar{g} \in \bar{R}$. Then $x - fg \in (xy - zw)$ unless $x - fg = 0$. Since $xy - zw$ is irreducible in $\mathbb{Q}[x, y, z, w]$, we must have $x \mid f$ or $x \mid g$. If $x \mid f$, then $\bar{f}$ is a unit in $\bar{R}$, so $\bar{x}$ is irreducible. Similarly, if $x \mid g$, then $\bar{g}$ is a unit in $\bar{R}$, so $\bar{x}$ is irreducible. Thus, $\bar{x}$ is irreducible. By symmetry, $\bar{y}, \bar{z}, \bar{w}$ are also irreducible. However, we have the two distinct factorizations $\bar{x}\bar{y} = \bar{z}\bar{w}$ into irreducibles, and the irreducibles $\bar{x}, \bar{y}, \bar{z}, \bar{w}$ are not associates of each other since they are all of degree one. Therefore, $\bar{R}$ is not a unique factorization domain.

Theorem 1.74. *Let R be a principal ideal domain. Then R is a unique factorization domain.*

Proof. Let $x \in R$ be a non-zero non-unit. We show that x can be written as a product of irreducibles. Let $\mathfrak{m}$ be a maximal ideal of R containing (x). Since R is a principal ideal domain, $\mathfrak{m} = (p_1)$ for some $p_1 \in R$. Thus, p_1 is a prime element, so p_1 is irreducible. If $x = p_1 x_1$ for some $x_1 \in R$, then we can repeat the process for x_1. If x_1 is a unit, then we are done. Else, we can repeat the process for x_1 to get a prime element p_2 such that $x_1 = p_2 x_2$ for some $x_2 \in R$. We can keep repeating this process; we claim that it must terminate after finitely many steps. If not, then we have an infinite sequence of ideals $(x) \subsetneq (x_1) \subsetneq (x_2) \subsetneq \cdots$. Let $I = \bigcup_{n \geq 0} (x_n)$. Then I is an ideal of R. Since R is a principal ideal domain, $I = (y)$ for some $y \in R$. Thus, $y \in I$ implies that $y \in (x_r)$ for some $r \geq 0$, so $y = ux_r$ for some $u \in R$. Also, $x_r = p_{r+1} x_{r+1}$ for some $p_{r+1} \in R$ and $x_{r+1} \in R$, so $y = up_{r+1} x_{r+1}$. Also, since $x_{r+1} = vy$ for some $v \in R$, we have $y = up_{r+1} vy$, so $(1 - up_{r+1} v)y = 0$. Since R is an integral domain and $y \neq 0$, we must have $up_{r+1} v = 1$, so p_{r+1} is a unit, contradicting the fact that p_{r+1} is a prime element. Therefore, the

process must terminate after finitely many steps, so x can be written as a product of irreducibles (note that the last terminating unit may be absorbed into the previous prime element to make it irreducible).

To show uniquness upto reordering, and associates, let $x = p_1 \cdots p_n = q_1 \cdots q_m$ for some $m \geq 1$ and irreducibles q_j. Since p_1 is irreducible and R is a principal ideal domain, p_1 is prime, so $p_1 \mid q_1 \cdots q_m$ implies that $p_1 \mid q_{i_1}$ for some $1 \leq i_1 \leq m$. Since q_{i_1} is irreducible, p_1 and q_{i_1} are associates, so $p_1 = u_1 q_{i_1}$ for some unit u_1. We can reorder the q_j's so that $i_1 = 1$, giving us $p_1 = u_1 q_1$. Cancellation gives us $p_2 \cdots p_n = u_1 q_2 \cdots q_m$; we can repeat the same process to get $p_2 = u_2 q_2$ for some unit u_2, and so on. After repeating this process, we get $p_i = u_i q_i$ for some unit u_i for all $1 \leq i \leq n$. By symmetry argument, we also get $q_j = v_j p_j$ for some unit v_j for all $1 \leq j \leq m$. Thus, $n = m$ and $p_i = u_i q_i$ for some unit u_i for all $1 \leq i \leq n$, so the factorization of x into irreducibles is unique up to reordering and up to associates. ■

Proposition 1.75. *Let R be a unique factorization domain. Then an element is irreducible if and only if it is prime.*

Proof. We already know that every prime element is irreducible in any integral domain. For the converse, let x be an irreducible element of R. Let $x \mid ab$ for some $a, b \in R$. There then exists some $y \in R$ such that $ab = xy$. Let $a = p_1 \cdots p_n$ and $b = q_1 \cdots q_m$, and $y = r_1 \cdots r_k$ be the factorizations of a, b, y into irreducibles. Then $p_1 \cdots p_n q_1 \cdots q_m = r_1 \cdots r_k x$. Since x is irreducible, x must be an associate of some p_i or some q_j, so $x \mid a$ or $x \mid b$. Thus, x is prime. ■

Theorem 1.76. *Let R be a principal ideal domain. Then R has a Dedekind-Hasse norm.*

Proof. Let $x \in R \setminus \{0\}$. Define $N : R \setminus \{0\} \to \mathbb{Z}_{\geq 0}$ as follows: if x is a unit, $N(x) = 1$. Otherwise, write $x = p_1 \cdots p_n$ for some $n \geq 1$ and irreducibles p_i, and define $N(x) = 2^n$. Verify that N is a Dedekind-Hasse norm. ■

February 19th.

Proposition 1.77. *Let R be a unique factorization domain. If $a, b \in R$ are non-zero, then $\gcd(a, b)$ exists. If a or b is a unit, then $\gcd(a, b)$ is a unit. Moreover, if a and b are non-units, then there exist irreducibles $p_1, \ldots, p_r$ such that $a = p_1^{e_1} \cdots p_r^{e_r}$ and $b = p_1^{f_1} \cdots p_r^{f_r}$ for some $e_i, f_i \geq 0$ for all $1 \leq i \leq r$, and $\gcd(a, b) = p_1^{\min\{e_1, f_1\}} \cdots p_r^{\min\{e_r, f_r\}}$.*

Proof. If a or b is a unit, then $\gcd(a, b)$ is a unit since any unit divides any element. So we may assume that a and b are non-units. Since R is a unique factorization domain, there exist irreducibles $p_1, \ldots, p_r$ such that $a = p_1^{e_1} \cdots p_r^{e_r}$ and $b = p_1^{f_1} \cdots p_r^{f_r}$ for some $e_i, f_i \geq 0$ for all $1 \leq i \leq r$. Let $d = p_1^{\min\{e_1, f_1\}} \cdots p_r^{\min\{e_r, f_r\}}$. Then $d \mid a$ and $d \mid b$. If $d' \in R$ is such that $d' \mid a$ and $d' \mid b$, then there exist some irreducibles $q_1, \ldots, q_s$ such that $d' = q_1^{g_1} \cdots q_s^{g_s}$ for some $g_i \geq 0$ for all $1 \leq i \leq s$. Since $d' \mid a$, we have that each irreducible factor of d' must be an associate of some irreducible factor of a, so after reordering, we can write $d' = p_1^{g_1'} \cdots p_r^{g_r'}$ where each $g_i' \geq 0$. Since $d' \mid b$, we have that each irreducible factor of d' must be an associate of some irreducible factor of b, so after reordering, we can write $d' = p_1^{g_1''} \cdots p_r^{g_r''}$ where each $g_i'' \geq 0$. Thus, after reordering, we can write $d' = p_1^{g_1'''} \cdots p_r^{g_r'''}$ where $g_i''' = \min\{g_i', g_i''\} \leq \min\{e_i, f_i\}$ for all $1 \leq i \leq r$. Thus, $d' \mid d$. Therefore, d is a gcd of a and b. ■

Proposition 1.78. *Let R be an integral domain. Then R is a unique factorization domain if and only if every irreducible element of R is prime and any increasing chain of principal ideals in R stabilizes.*

Proof. For the foward implication, if have already shown that every irreducible element of R is prime. Let $(x_1) \subsetneq (x_2) \subsetneq (x_3) \subsetneq \cdots$ be a strictly increasing chain of principal ideals in R. If x_1 is a unit, then $(x_1) = R$, contradicting the fact that $(x_1) \subsetneq (x_2)$. If $x_1 = 0$, then $x_2 \neq 0$ so we may assume that x_1 is a non-zero non-unit. Since R is a unique factorization domain, we can uniquely factor x_1 as $x_1 = p_1 \cdots p_n$ for some $n \geq 1$ and irreducibles p_i. Since $x_1 \in (x_2)$, $x_1 = x_2 y$ for some $y \in R$. Also $(x_1) \subsetneq (x_2)$ implies that y is not a unit. Let $x_2 = q_1 \cdots q_m$ and $y = r_1 \cdots r_k$ be the factorizations of x_2 and y into irreducibles. Then by the uniquness of factorization, $n = m + k$ and since y is not a unit, $k \geq 1$, so $m < n$. Thus, x_2 has strictly less irreducible factors than x_1. We can repeat the same process for x_2 to get that x_3 has

strictly less irreducible factors than x_2, and so on. Since the number of irreducible factors is a non-negative integer, this process must terminate after finitely many steps giving a x_n that is a unit, contradicting the fact that $(x_n) \subsetneq (x_{n+1})$. Therefore, any increasing chain of principal ideals in R stabilizes.

For the converse implication, let $x \in R$ be a non-zero non-unit. The result is trivial if x is irreducible, so we may assume that x is not an irreducible. We claim that there exists an irreducible $p_1 \in R$ such that $x = p_1 y$ for some $y \in R$. Since x is not irreducible, there exist $x_1, y_1 \in R$ such that $x = x_1 y_1$ and neither x_1 nor y_1 is a unit. If x_1 is irreducible, then we can take $p_1 = x_1$ and $y = y_1$. Else, we can repeat the same process for x_1 to get $x = x_2 y_2 y_1$ for some non-units $x_2, y_2 \in R$. We can keep repeating this process; we claim that it must terminate after finitely many steps. If not, then we have an infinite sequence of ideals $(x) \subsetneq (x_1) \subsetneq (x_2) \subsetneq \cdots$. This contradicts the fact that any increasing chain of principal ideals in R stabilizes. Therefore, the process must terminate after finitely many steps, so there exists an irreducible $p_1 \in R$ such that $x = p_1 y$ for some $y \in R$. If y is a unit, then we are done. Else, we can repeat the same process for y to get $x = p_1 p_2 y_2$ for some irreducible p_2 and some $y_2 \in R$. We can keep repeating this process; we claim that it must terminate after finitely many steps. If not, then we have an infinite sequence of ideals $(x) \subsetneq (p_1) \subsetneq (p_1 p_2) \subsetneq \cdots$. This contradicts the fact that any increasing chain of principal ideals in R stabilizes. Therefore, the process must terminate after finitely many steps, so x can be written as a product of irreducibles. The uniquenss follows from the same proof as that of the theorem that a principal ideal domain is a unique factorization domain. ■

1.8 Localization

March 2nd.

Localization of rings is a process that allows us to 'invert' a subset of elements in a ring. We will see that the localization of $\mathbb{Z}$ leads to $\mathbb{Q}$.

Definition 1.79. Let R be an integral domain. Then $K := \{(a,b) \equiv \frac{a}{b} \mid a, b \in R,\ b \neq 0\} = R \times (R \setminus \{0\})/\sim$ where $(a,b) \sim (c,d)$ if and only if $ad = bc$ is a field, called the *field of fractions* of R; the operations in this field are defined as follows: $\frac{a}{b} + \frac{c}{d} = \frac{ad+bc}{bd}$ and $\frac{a}{b} \cdot \frac{c}{d} = \frac{ac}{bd}$.

We may denote the field of fractions of R by $QF(R)$ or $\mathrm{frac}(R)$. Note that the field of fractions of $\mathbb{Z}$ is $\mathbb{Q}$; the definition is simply abstracted from the construction of $\mathbb{Q}$ from $\mathbb{Z}$. Note that we have a natural embedding $R \hookrightarrow K$ given by $a \mapsto \frac{a}{1}$ for all $a \in R$. We will identify R with its image in K under this embedding, so we will view R as a subring of K. If R itself is a field, then R is its own field of fractions. We wish to generalize this idea; that is, we want to 'invert' only a particular subset of elements in R instead of all non-zero elements.

Definition 1.80. Let R be a ring. A subset $S \subseteq R$ is called a *multiplicative subset* if $1 \in S$ and if $s, t \in S$, then $st \in S$.

For example, if R is a ring then any subring S works as a multiplicative subset. If $R = \mathbb{Z}$, we can take $S = \mathbb{N}$. We can also take S to be $R^\times$, the group of units. If R is an integral domain and $S = R \setminus \{0\}$, then S is a multiplicative subset. For any R, we can also take $S = I \setminus \{0\} \cup \{1\}$ for any ideal I of R. In particular, we are interested in this example: if R is a ring and P is a prime ideal of R, then $S = R \setminus P$ is a multiplicative subset. To show this, let $x, y \in R \setminus P$. If $xy \in P$, then since P is prime, either $x \in P$ or $y \in P$, contradicting the choice of x, y. Thus, $xy \in R \setminus P$. Moreover, P is a proper ideal, so $1 \in R \setminus P$. Thus, $R \setminus P$ is a multiplicative subset. Similarly, $S = R \setminus \bigcup_{i=1}^n P_i$ is a multiplicative subset for any prime ideals $P_1, \ldots, P_n$ of R.

Definition 1.81. Let R be a ring and $S \subseteq R$ be a multiplicative subset. Then on $S \times R$, we define an (equivalence) relation $\sim$ by $(s_1, r_1) \sim (s_2, r_2)$ if and only if there exists some $t \in S$ such that $t(s_1 r_2 - s_2 r_1) = 0$. We can verify that $\sim$ is an equivalence relation. Let $S^{-1}R = S \times R/\sim$. We define $\oplus$ and $\odot$ on $S^{-1}R$ as follows: $\frac{r_1}{s_1} \oplus \frac{r_2}{s_2} = \frac{s_2 r_1 + s_1 r_2}{s_1 s_2}$ and $\frac{r_1}{s_1} \odot \frac{r_2}{s_2} = \frac{r_1 r_2}{s_1 s_2}$. These operations are well-defined, and $S^{-1}R$ is a ring with respect to these operations, called the *localization* of R at S. Finally, we have a natural homomorphism $\varphi : R \to S^{-1}R$ defined by $\varphi(r) = \frac{r}{1}$ for all $r \in R$.

March 5th.

We will simply use $+$ and $\cdot$ for $\oplus$ and $\odot$ respectively. We verify some of the claims made in this definition.

Proof. That $\sim$ is reflexive and symmetric is clear. To show that $\sim$ is transitive, let $(s_1, r_1) \sim (s_2, r_2)$ and $(s_2, r_2) \sim (s_3, r_3)$. Then there exist some $t, t' \in S$ such that $t(s_1r_2 - s_2r_1) = 0$ and $t'(s_2r_3 - s_3r_2) = 0$. Then, multiplying the first equation by s_3t' and the second equation by s_1t, we get $ts_3t'(s_1r_2 - s_2r_1) = 0$ and $s_1tt'(s_2r_3 - s_3r_2) = 0$. Adding these two equations gives us $ts_3t'(s_1r_2 - s_2r_1) + s_1tt'(s_2r_3 - s_3r_2) = 0$, so $tt'(s_1s_3r_2 - s_1s_2r_3 + s_1s_2r_3 - s_1s_3r_2) = 0$, so $tt'(s_1s_3r_2 - s_1s_3r_2) = 0$, so $tt' \cdot 0 = 0$. Thus, $(s_1, r_1) \sim (s_3, r_3)$, so $\sim$ is transitive. $\sim$ is an equivalence relation.

To show that $\odot$ is well-defined, let $\frac{r_1}{s_1} = \frac{r_1'}{s_1'}$ and $\frac{r_2}{s_2} = \frac{r_2'}{s_2'}$. Then there exist some $u, v \in S$ such that $u(s_1'r_1 - s_1r_1') = 0$ and $v(s_2'r_2 - s_2r_2') = 0$. Then $uv(s_1's_2'r_1r_2 - s_1s_2r_1r_2') = uv(s_1'r_1 \cdot s_2'r_2 - s_1r_1' \cdot s_2r_2') = uv(s_1'r_1 \cdot s_2'r_2 - s_1'r_1 \cdot s_2r_2' + s_1'r_1 \cdot s_2r_2' - s_1r_1' \cdot s_2r_2') = uv(s_1'r_1 \cdot (s_2'r_2' - s_2'r_2') + (s_1'r_1' - s_1'r_1') \cdot s_2'r_2') = uv(0+0) = 0$. Thus, $\frac{r_1r_2}{s_1s_2} = \frac{r_1'r_2'}{s_1's_2'}$, so $\odot$ is well-defined. Similarly, we can show that $\oplus$ is well-defined.

Showing that $(S^{-1}R, \oplus)$ is an abelian group is straightforward, with the identity element being $\frac{0}{1}$ and the inverse of $\frac{r}{s}$ being $\frac{-r}{s}$. The distributivity of $\odot$ over $\oplus$, and its associativity and commutativity, are also straightforward to verify, with $\frac{1}{1}$ being the multiplicative identity. Thus, $S^{-1}R$ is a ring with respect to these operations. Finally, φ may be verified to be a homomorphism by checking that $\varphi(r_1 + r_2) = \varphi(r_1) \oplus \varphi(r_2)$ and $\varphi(r_1r_2) = \varphi(r_1) \odot \varphi(r_2)$ for all $r_1, r_2 \in R$. ■

Example 1.82. The most common example, of course, is to take $\mathbb{R} = \mathbb{Z}$ and $S = \mathbb{Z} \setminus \{0\}$. Then $S^{-1}R$ is the field of fractions of $\mathbb{Z}$, which is $\mathbb{Q}$. More generally, if R is an integral domain and $S = R \setminus \{0\}$, then $S^{-1}R$ is the field of fractions of R. In such a case, the natural homomorphism $\varphi : R \to S^{-1}R$ is injective, so we can identify R with its image in $S^{-1}R$, and view R as a subring of $S^{-1}R$. Thus, we can view the field of fractions of an integral domain as an extension of the integral domain.

Proposition 1.83. *Let $\varphi : R \to S^{-1}R$ be the natural homomorphism. Then $\ker \varphi = \{r \in R \mid sr = 0$ for some $s \in S\}$. In particular, if S consists only of non-zero divisors, then φ is injective.*

Proof. The second statement follows from the first statement since $\ker \varphi = \{0\}$ if S consists only of non-zero divisors. To show the first statement, let $r \in R$. If $r \in \ker \varphi$, then $\varphi(r) = \frac{r}{1} = 0$ in $S^{-1}R$, so there exists some $s \in S$ such that $s(r \cdot 1 - 0 \cdot 1) = sr = 0$. Conversely, if $sr = 0$ for some $s \in S$, then $\varphi(r) = \frac{r}{1} = \frac{0}{s} = 0$ in $S^{-1}R$, so $r \in \ker \varphi$. Thus, $\ker \varphi = \{r \in R \mid sr = 0 \text{ for some } s \in S\}$. ■

From here, we verify that $S^{-1}R$ indeed gives the field of fractions for an integral domain R by verifying that it is a field; indeed; let $\frac{r}{s} \in S^{-1}R$ such that $\frac{r}{s} \neq \frac{0}{1}$; this occurs iff $t(r - 0) \neq 0$ for all $t \in S$ which occurs iff $r \neq 0$, since R is an integral domain. This means that $r \in S$ so $\frac{s}{r} \in S^{-1}R$ showing that $\frac{r}{s}$ has a multiplicative inverse.

An important example of a multiplicative subset of a ring R is to choose $S = \{1, x, x^2, x^3, \ldots\}$ for some $x \in R$. In such a case, we denote $S^{-1}R = R[\frac{1}{x}]$. In particular, if we choose $R = \mathbb{Z}$ and $S = \{1, 2, 4, 8, \ldots\}$, then we get $S^{-1}R = \mathbb{Z}[\frac{1}{2}]$. If R is an integral domain, we can write $R[\frac{1}{x}]$ as $\{\frac{a}{x^k} \mid a \in R,\ k \geq 0\}$. Some choices of S can lead to rather ugly constructions; suppose $0 \in S$ for the ring R. Then $\frac{r_1}{s_1}, \frac{r_2}{s_2} \in S^{-1}R$ are always equal; $0 \cdot (s_1r_2 - s_2r_1) = 0$ since $0 \in S$, making the construction a trivial ring.

Proposition 1.84. *Let R be a ring, S a multiplicative subset, and $\varphi : R \to S^{-1}R$ be the localization homomorphism. Then for $s \in S$, $\varphi(s)$ is a unit in $S^{-1}R$.*

Proof. s is mapped to $\frac{s}{1}$. Since $s \in S$, $\frac{1}{s} \in S^{-1}R$, which is the multiplicative inverse of $\frac{s}{1}$; $\varphi(s)$ is a unit. ■

Let R be an integral domain, and let $S_x = \{1, x, x^2, \ldots\}$ and $S_y = \{1, y, y^2, \ldots\}$ for some $x, y \in R$. Also let $S_{xy} = \{1, xy, x^2y^2, \ldots\}$. Then, we have $S_x^{-1}R \subseteq S_{xy}^{-1}R$.

March 6th.

Recall that we define the quotient map as follows: given any ring homomorphism $\varphi : R \to A$ such that $I \subseteq \ker \varphi$ for I an ideal of R, then there exists a unique map $\bar{\varphi} : R/I \to A$ such that $\bar{\varphi} \circ q = \varphi$, via $\bar{\varphi}(r + I) = \varphi(r)$. One can also define the quotient ring as follows.

Definition 1.85. A quotient of a ring R by an ideal I is a ring homomorphism $q : R \to B$ such that for any ring homomorphism $\varphi : R \to A$ with $I \subseteq \ker \varphi$, there exists a unique ring homomorphism $\bar{\varphi} : B \to A$ satisfying $\bar{\varphi} \circ q = \varphi$.

Theorem 1.86 (The *universal property of localization*)**.** *Let R be a ring and S a multiplicative subset. Also let $f : R \to A$ be a ring homomorphism such that $f(s)$ is a unit in A for all $s \in S$. Then there exists a unique homomorphism $\bar{f} : S^{-1}R \to A$ such that $\bar{f} \circ \varphi = f$, where $\varphi : R \to S^{-1}R$ is the localization map.*

Proof. We define $\bar{f} : S^{-1}R \to A$ as $\bar{f}(\frac{r}{s}) = f(s)^{-1}f(r)$. This $\bar{f}$ is well-defined; to show this let $\frac{r}{s} = \frac{r'}{s'}$. That is, there exists $u \in S$ such that $u(s'r - sr') = 0$. Thus,

$$f(u)(f(s')f(r) - f(s)f(r')) = 0 \implies f(s')f(r) = f(s)f(r') \implies \bar{f}(\frac{r}{s}) = \bar{f}(\frac{r'}{s'}) \tag{1.48}$$

since $u, s, s' \in S$ are all mapped to units in A. To show $\bar{f}$ is a ring homomorphism, we first have

$$\bar{f}(\frac{r_1}{s_1} + \frac{r_2}{s_2}) = \bar{f}(s_1s_2)^{-1}f(s_2r_1 + s_1r_2) = f(s_1)^{-1}f(r_1) + f(s_2)^{-1}f(r_2) = \bar{f}(\frac{r_1}{s_1}) + \bar{f}(\frac{r_2}{s_2}). \tag{1.49}$$

Similarly, one can show multiplication. For uniqueness, let $g : S^{-1}R \to A$ be a ring homomorphism such that $g \circ \varphi = f$. Then $g(\frac{r}{s}) = g(\frac{r}{1})g(\frac{1}{s}) = g \circ \varphi(r)(g \circ \varphi(s))^{-1} = f(r)f(s)^{-1} = \bar{f}(\frac{r}{s})$ showing $g = \bar{f}$. ■

We look at some applications of localization.

Proposition 1.87. *Let R be an integral domain and K be a field. Moreover, let $i : R \to K$ be an injective ring homomorphism. Then i extends to an injective ring homomorphism $\tilde{\iota} : \mathrm{frac}(R) \to K$. In other words, if $R \subseteq K$, then $\mathrm{frac}(R) \subseteq K$.*

Here, recall that $\mathrm{frac}(R)$ is the field of fractions of an integral domain R; its the localization of R with $R \setminus \{0\}$ chosen as the multiplicative subset.

Proof. Note that $i(s)$ is a unit in K for all $s \neq 0$. By the universal property, there exists a unique ring homomorphism $\tilde{\iota} : \mathrm{frac}(R) \to K$ such that $\tilde{\iota} \circ \varphi = i$, with φ being the localization map. Thus, $\tilde{\iota}(\frac{r}{1}) = i(r)$ for all $r \in R$. Moreover, $\tilde{\iota} : \mathrm{frac}(R) \to K$ is also injective; we have

$$\tilde{\iota}(\frac{r}{s}) = 0 \implies \tilde{\iota}(\frac{r}{1}) = i(r) = 0 \implies r = 0 \implies \frac{r}{s} = \frac{0}{1} \tag{1.50}$$

since i is injective. ■

Proposition 1.88. *Let R be a ring and $a \in R$. Then,*

$$R[X]/(aX - 1) \cong R[\frac{1}{a}] \tag{1.51}$$

via the map $r \mapsto \frac{r}{1}$ for all $r \in R$ and $X \mapsto \frac{1}{a}$.

Proof. Define the map $\alpha : R[X] \to S^{-1}R = R[\frac{1}{a}]$ via $f(X) \mapsto f(\frac{1}{a})$. Let $I = (aX - 1)$. Note that $\alpha(aX - 1) = \frac{a}{1} \cdot \frac{1}{a} - \frac{1}{1} = 0$, so $(aX - 1) \subseteq \ker \alpha$. Thus, we have a ring homomorphism $\bar{\alpha} : R[X]/I \to S^{-1}R$ such that $\bar{\alpha} \circ q = \alpha$, q being the quotient map; $\bar{\alpha}(f(X) + I) = f(\frac{1}{a})$. We now give a map in the other direction; let $f : R \to R[X]/I$ be the map defined by $f(r) = r + I$. Note that $S = \{1, a, a^2, \ldots\}$; each a^n is mapped to the unit $a^n + I$ with inverse $X^n + I$. Thus, by the universal property of localization, there exists a unique ring homomorphism $\bar{f} : S^{-1}R \to R[X]/(aX - 1)$ such that $\bar{f} \circ \varphi = f$. We show that $\bar{\alpha}$ and $\bar{f}$ are inverses of each other. To this end, see that

$$\bar{f} \circ \bar{\alpha}(g(X) + I) = \bar{f} \circ \alpha(g(X)) = \bar{f} \circ g(\frac{1}{a}) = (b_n + I)\bar{f}(\frac{1}{a^n}) + (b_{n-1} + I)\bar{f}(\frac{1}{a^{n-1}}) + \cdots (b_0 + I) \tag{1.52}$$

which is $(b_n + I)(X^n + I) + \cdots + (b_1 + I)(X + I) + b_0 + I = (b_nX^n + b_{n-1}X^{n-1} + \cdots + b_0) + I = g(X) + I$. Similarly, one can show that the $\bar{\alpha} \circ \bar{f}$. ■

1.8.1 Local Ring

Definition 1.89. A ring is called a *local ring* if it has a unique maximal ideal.

Of course, a trivial example is fields; the unique maximal ideal is $\{0\}$. In the ring of power series $k[[X]]$, (X) is the unique maximal ideal making $k[[X]]$ a local ring for a field k. $\mathbb{Z}/p^r\mathbb{Z}$ is a local ring with $p\mathbb{Z}/p^r\mathbb{Z}$ the unique maximal ideal.

For a more particular example, let $R \subseteq \mathbb{C}(X)$ be the subring of the field of complex rational functions defined as follows: $R = \{f(X)/g(X) \mid f(X), g(X) \in \mathbb{C}[X] \text{ and } g(0) \neq 0\}$. One can verify that this is a ring; it is the ring of rational functions from $\mathbb{C}$ to $\mathbb{C}$ which is defined at 0. Since such a function is continuous in its domain, it is locally define at 0. In fact, R is a local ring with the maximal ideal consisting of those rational functions h such that $h(0) = 0$.

March 12th.

Proposition 1.90. *Let R be a ring and P be a prime ideal of R. Set $S = R \setminus P$. Then $S^{-1}R$ is a local ring with unique maximal ideal $PS^{-1}R = \{\frac{r}{s} \mid r \in P,\ s \in R \setminus P\}$.*

Proof. Note that it is enough to show that $PS^{-1}R$ is an ideal and $\alpha \in S^{-1}R \setminus PS^{-1}R$ is a unit. To show that $PS^{-1}R$ is an ideal, let $\frac{r_1}{s_1}, \frac{r_2}{s_2} \in PS^{-1}R$, so that $r_1, r_2 \in P$ and $s_1, s_2 \in R \setminus P$. Then $s_2r_1 + s_1r_2 \in P$ since P is and ideal, and $s_1s_2 \in R \setminus P$. Thus, $\frac{s_2r_1+s_1r_2}{s_1s_2} \in PS^{-1}R$. Also if $\frac{R}{s} \in S^{-1}R$, then $\frac{rr_1}{ss_1} \in PS^{-1}R$. Hence $PS^{-1}R$ is an ideal of $S^{-1}R$. Now let $\alpha = \frac{r}{s} \in S^{-1}R \setminus PS^{-1}R$. Since $\alpha \notin PS^{-1}R$, $r \notin P$. This means that $r \in R \setminus P = S$, so $\frac{s}{r} \in S^{-1}R$ is the multiplicative inverse of α. Thus, $S^{-1}R$ is a local ring with unique maximal ideal $PS^{-1}R$. ■

Proposition 1.91. *Let R be a ring $\mathfrak{m}$ be a maximal ideal of R. Then the following are equivalent:*

(i) R is a local ring.

(ii) The set of non-units of R form the ideal $\mathfrak{m}$.

(iii) $1 + x$ is a unit for all $x \in \mathfrak{m}$.

Proof. $(i) \implies (ii)$: Let J be the set of non-units of R. Note that $\mathfrak{m} \subseteq J$ since $\mathfrak{m}$ is a proper ideal. Let $x \in J$. Since x is not a unit, there exists a maximal ideal $\mathfrak{m}'$ containing x. Since R is a local ring, $\mathfrak{m}' = \mathfrak{m}$, so $x \in \mathfrak{m}$. Thus, $J \subseteq \mathfrak{m}$, so $J = \mathfrak{m}$.

$(ii) \implies (iii)$: If $x \in \mathfrak{m}$ and $1 + x$ is not a unit, then $1 + x \in \mathfrak{m}$. But then $1 = (1 + x) - x \in \mathfrak{m}$, contradicting the fact that $\mathfrak{m}$ is a proper ideal. Thus, $1 + x$ is a unit for all $x \in \mathfrak{m}$.

$(iii) \implies (i)$: Let $x \in \mathfrak{m}$. Then $ax \in \mathfrak{m}$ for all $a \in R$, and thus $1 + ax$ is a unit in R for all $a \in R$. Hence, $x \in \mathrm{Jac}(R)$. Finally, we have $\mathfrak{m} \subseteq \mathrm{Jac}(R)$ and so, $\mathfrak{m} = \mathrm{Jac}(R)$ since $\mathrm{Jac}(R) \subseteq \mathfrak{m}$ for any maximal ideal $\mathfrak{m}$. Thus, R is a local ring. ■

We look at some examples of using these propositions to construct local rings.

Example 1.92. Let $R = \mathbb{Z}$ and $P = (3)$. Then $S = \mathbb{Z} \setminus (3) = \{1, 2, 4, 5, 7, 8, \ldots\}$ and $S^{-1}R$ is the ring $\mathbb{Z}_{(3)} = \{\frac{a}{b} \mid a, b \in \mathbb{Z},\ 3 \nmid b\}$, which is a local ring with unique maximal ideal $\{\frac{a}{b} \mid a, b \in \mathbb{Z},\ 3 \nmid b,\ 3|a\}$.

1.9 Polynomials and Gauss Lemma

March 13th.

Gauss lemma can refer to several results; the one Gauss originally proved is the following: let $f(X) \in \mathbb{Z}[X]$ be a polynomial such that $p \nmid f(X)$ for all primes p. If $f(X) = g(X)h(X)$ in $\mathbb{Q}[X]$ then there exists a factorization $f(X) = G(X)H(X)$ in $\mathbb{Z}[X]$ such that $\deg(g) = \deg(G)$ and $\deg(h) = \deg(H)$. We will see this result as a corollary of several sequence of results, which are also called Gauss lemma.

Definition 1.93. Let R be a ring which admits a gcd function (for example, a UDF). For $f(X) \in R[X]$, we define the *content* of $f(X)$ as $c(f) = \gcd(a_0, a_1, \ldots, a_n)$ where $f(X) = a_n X^n + \cdots + a_1 X + a_0$. Note that $c = c(f)$ holds if and only if $uc = c(f)$ for all units $u \in R$. A polynomial $f(X)$ is called *primitive* if $c(f) = 1$.

As an example, $2X - 1$ is a primitive polynomial in $\mathbb{Z}[X]$ since $c(2X - 1) = 1$. On the other hand, $2X + 2$ is not a primitive polynomial in $\mathbb{Z}[X]$ since $c(2X + 2) = 2$. Note that if $f(X) \in R[X]$ is a non-zero polynomial, then $f(X) = c(f)f_0(X)$ for some primitive polynomial $f_0(X)$; indeed, we can take $f_0(X) = \frac{1}{c(f)} f(X)$.

The first version of Gauss lemma is the following.

Lemma 1.94 (Gauss lemma, version 1)**.** *Let R be a UFD and $f(x), g(x) \in R[X]$. Then $c(fg) = c(f)c(g)$.*

Lemma 1.95 (Gauss lemma, version 2)**.** *Let R be a UFD and $K = \mathrm{frac}(R)$ be its field of fractions. Let $f(X) \in R[X] \subseteq K[X]$. If $f(X) = g(X)h(X)$ for some $g(X), h(X) \in K[X]$, then $f(X)$ can be factored as $f(X) = G(X)H(X)$ for some $G(X), H(X) \in R[X]$ such that $\deg(G) = \deg(g)$ and $\deg(H) = \deg(h)$.*

One can easily see that the statement at the beginning of this section is a corollary of the second version of Gauss lemma. The following version of Gauss lemma, a corollary of the second version, is the most widely used version of Gauss lemma.

Corollary 1.96 (Gauss lemma, version 3)**.** *Let R be a UFD and $K = \mathrm{frac}(R)$ be its field of fractions. A primitive polynomial $f(X) \in R[X]$ is irreducible in $R[X]$ if and only if it is irreducible in $K[X]$.*

We first begin with the proof of this corollary.

Proof of Gauss lemma, version 3 from version 2. For the forward implication, let $f(X) \in R[X]$ be a primitive polynomial which is irreducible in $R[X]$, and suppose $f(X) = g(X)h(X)$ for some $g(X), h(X) \in K[X]$. By version 2 of Gauss lemma, there exist some $G(X), H(X) \in R[X]$ such that $f(X) = G(X)H(X)$ and $\deg(G) = \deg(g) \geq 1$ and $\deg(H) = \deg(h) \geq 1$. Hence, G and H are non-units in $R[X]$, contradicting the irreducibility of $f(X)$ in $R[X]$. For the converse implication, assume $f(X)$ is irreducible in $K[X]$, and suppose $f(X) = G(X)H(X)$ for some $G(X), H(X) \in R[X]$. Since $f(X)$ is irreducible in $K[X]$, either $G(X)$ or $H(X)$ is a constant polynomial; let's say $G(X)$ is a constant polynomial. Then $f(X) = cH(X)$ for some $c \in R$. Since $f(X)$ is primitive, we must have $c \mid c(f) = 1$, so c is a unit in R. Thus, $f(X)$ is irreducible in $R[X]$. ■

We now prove the second verion.

Proof of Gauss lemma, version 2 from version 1. Let $f(X) \in R[X]$ be a polynomial such that $f(X) = g(X)h(X)$ with $g(X), h(X) \in K[X]$. Note that since each coefficient of $g(X)$ and $h(X)$ is a fraction, we can find some β (possibly, the product of all denominators of coefficients of $g(X)$) and α (similarly pick α) such that $G_1(X) = \beta g(X)$ and $H_1(X) = \alpha h(X)$ are in $R[X]$. Then $\alpha\beta f(X) = G_1(X)H_1(X)$, so $c(\alpha\beta f(X)) = \alpha\beta c(f) = c(G_1)c(H_1)$. So $f(X) = c(f)F(X)$ where $F(X) \in R[X]$. Similarly, $G_1(X) = c(G_1)G(X)$ and $H_1(X) = c(H_1)H(X)$ for some $G, H \in R[X]$. Thus, $c(f)F(X) = c(G_1)c(H_1)G(X)H(X)$, so $F(X) = uG(X)H(X)$ where $u \in R$ is a unit. Hence, $uc(f)G(X), H(X) \in R[X]$ and $f(X) = c(F)F(X)$, with $\deg(uc(f)G(X)) = \deg(G(X)) = \deg(g(X))$ and $\deg(H(X)) = \deg(h(X))$. ■

Finally, we prove the first version of Gauss lemma.

Proof of Gauss lemma, version 1. Let $f(X), g(X) \in R[X]$, and $h(X) = f(X)g(X)$. Let $f(X) = c(f)F(X)$, $g(X) = c(g)G(X)$, $h(X) = c(h)H(X)$ where $F(X), G(X), H(X) \in R[X]$. Then $c(h)H(X) = c(f)c(g)F(X)G(X)$, so Note that $c(f)c(g)$ divides all coefficients of $h(X)$, so $c(f)c(g) \mid c(h)$. Let $c(f)c(g) = p_1 \cdots p_n$ where p_i's are irreducible in R and $c(h) = p_1 \cdots p_n q_1 \cdots q_m$ where q_j's are irreducible in R, and $m, n \geq 0$. If $m \neq 0$, then q_1 exists and $q_1 \mid c(h)$, so that $p_1 \cdots p_n q_1 \mid c(h)H(X) = c(f)c(g)F(X)G(X)$. Thus $q_1 \mid F(X)G(X)$ since $c(f)c(g) = p_1 \cdots p_n$. Let i_0 be the smallest index i such that $q_1 \nmid b_{i_0}$ and let j_0 be

the smallest index j such that $q_1 \nmid c_{j_0}$. Now look at the coefficient a of $X^{i_0+j_0}$ in $F(X)G(X)$. Then $q_1 \mid a$. But $a = b_{i_0}c_{j_0} + b_{i_0-1}c_{j_0+1} + \cdots + b_0c_{j_0+i_0} + b_{i_0+1}c_{j_0-1} + \cdots b_{i_0+j_0}c_0 = b_{i_0}c_{j_0} +$ (terms divisible by q_1). Since $q_1 \nmid b_{i_0}$ and $q_1 \nmid c_{j_0}$, $q_1 \nmid b_{i_0}c_{j_0}$, so $q_1 \nmid a$, a contradiction since q_1 is a prime. Thus, $m = 0$, so $c(h) = c(f)c(g)$. ■

Example 1.97. Let $f(X,Y) \in \mathbb{C}[X,Y]$ be $f(X,Y) = 2X^3Y + 3X^2 + 7XY + 4$. Then $f(X,Y)$ is also in $\mathbb{C}(X)[Y]$, where $\mathbb{C}(X)$ is the formal power series field in X over $\mathbb{C}$. Note that $f(X,Y) = (2X^3 + 7X)Y + (3X^2 + 4)$ is irreducible in $\mathbb{C}(X)[Y]$ since it is linear in Y. One can verify that $\gcd(2X^3 + 7X, 3X^2 + 4) = 1$ so $f(X,Y)$ is primitive in $\mathbb{C}[X][Y]$. By Gauss lemma, $f(X,Y)$ is irreducible in $\mathbb{C}[X][Y] = \mathbb{C}[X,Y]$.

Theorem 1.98. *Let R be a UFD. Then $R[X]$ is also a UFD.*

Proof. Let $f(X) \in R[X]$, and write $f(X) = c(f)F(X)$ for some primitive $F(X) \in R[X]$ and $c(f) \in R$. Since R is a UFD, we have $c(f) = p_1 \cdots p_r$ for some irreducibles $p_i \in R$. Let $K = \mathrm{frac}(R)$. Then $F(X) \in K[X]$. But $K[X]$ is a euclidean domain, and hence a UFD; hence, $F(X) = g_1(X) \cdots g_n(X)$ where $g_i(X) \in K[X]$ are irreducible polynomials. By the second version of Gauss lemma, $F(X) = G_1(X) \cdots G_n(X)$ where $G_i(X) \in R[X]$, and $G_i = \alpha_i g_i$ for some $\alpha_i \in K^\times$. Also, $1 = c(F) = c(G_1) \cdots c(G_n)$ implies G_i's are primitive polynomials and, so, irreducile polynomials in $K[X]$. Therefore, $G_i(X)$ are irreducible in $R[X]$ by version 3 of Gauss lemma. So $f(X) = p_1 \cdots p_r G_1(X) \cdots G_n(X)$. Suppose $f(X) = q_! \cdots q_t$ for q_i irreduciles in $R[X]$. Note that $c(f) = c(q_1) \cdots c(q_t)$ by the first version of Gauss lemma. Let $q_i(X) = c(q_i)Q_i(X)$ where $Q_i(X) \in R[x]$ are primitive. Since $q_1(X)$ is irreducible, either $c(q_i)$ or $Q_i(X)$ is a unit, that is, $q_i(X)$ is primitive or $q_i(X)$ is a constant polynomial. After reordering, we may assume $q_1, \ldots, q_s$ are constant polynomials and $q_{s+1}(X), \ldots, q_t(X)$ are non-constant primitive polynomials. So $c(f) = c(q_1) \cdots c(q_r) = q_1 \cdots q_s$ and q_i are irreducible constants in $R[X]$ and hence irreducible in R. Since R is a UFD, $r = s$ and p_i's are associate to q_i's after reordering. Thus, $f(X) = c(f)F(X) = q_{s+1}(X) \cdots q_t(X)(q_1 \cdots q_s)$ so $F(X) = q_{s+1}(X) \cdots q_t(X) = G_1(X) \cdots G_n(X)$ in $R[X] \subseteq K[X]$. Since $K[X]$ is a UFD, after reordering, $n = t - s$ and $G_i(X) = u_i q_{s+i}(X)$ for some units $u_i \in K^\times$. Of course, these may not be units in R; let $u_i = \frac{a_i}{b_i}$ for some $a_i, b_i \in R$ with $b_i \neq 0$. Then $b_i G_i(X) = a_i q_{s+i}(X)$ which implies $c(B_i G_i(X)) = c(a_i q_{s+i}(X))$ showing a_i and b_i are associates, or that $\frac{a_i}{b_i}$ is a unit in R. ■

Index

associates, 17
associativity, 1

chinese remainder theorem, 13
commutative ring, 1
content, 23

Dedekind-Hasse, 16
degree, 3
distributivity, 1

euclidean domain, 14

field, 1
field of fractions, 19
first isomorphism theorem, 7

generating set, 5
greatest common divisor, 15
group ring, 6

ideal, 4
idempotent, 13
inclusion map, 12
integral domain, 6
irreducible element, 15
isomorphic, 2

Jacobson radical, 11

kernel, 5

local ring, 22
localization, 19

maximal ideal, 10
multiplicative subset, 19

nilpotent, 6
nilradical, 11

polynomial ring, 3
polynomials, 3
prime element, 15
prime ideal, 9
primitive, 23
principal ideal domain, 14
product ring, 12
projection map, 12

quotient map, 7
quotient ring, 7

reduced ring, 6
ring, 1
ring epimorphism, 2
ring homomorphism, 2
ring isomorphism, 2
ring monomorphism, 2
ring with unity, 1

second isomorphism theorem, 8
subring, 2

third isomorphism theorem, 8

unique factorization domain, 17
unit, 1
universal property of localization, 21
universal side divisor, 17

valuation map, 14
valuation ring, 14

zero divisor, 6